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Tissue resident macrophages specifically express *Lactotransferrin* and *Vegfc* during ear pinna regeneration in spiny mice

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Summary

The details of how macrophages control different healing trajectories (regeneration vs. scar formation) remain poorly defined. Spiny mice (*Acomys spp.*) can regenerate external ear pinnae tissue whereas lab mice (*Mus musculus*) form scar tissue in response to an identical injury. Here, we used this dual species system to dissect macrophage phenotypes between healing modes. We identified secreted factors from activated *Acomys* macrophages that induce a pro-regenerative phenotype in fibroblasts from both species. Transcriptional profiling of *Acomys* macrophages and subsequent *in vitro* tests identified VEGFC, PDGF-A and LTF as potential pro-regenerative modulators. Examining macrophages *in vivo*, we found that that *Acomys* resident macrophages secreted VEGFC and LTF whereas *Mus* macrophages do not. Lastly, we demonstrate the requirement for VEGFC during regeneration and find that interrupting lymphangiogenesis

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Declaration of Interest

All authors declare no competing interests except that at the time of publication, DT is now employed by Glaxosmithkline (GSK).

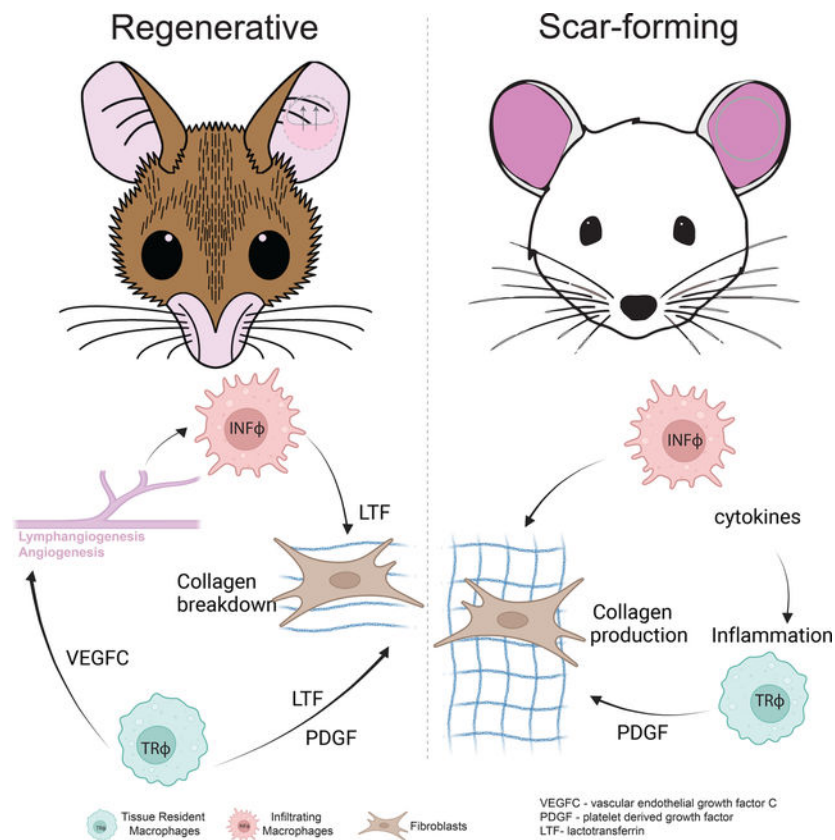
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delays blastema and new tissue formation. Together, our results demonstrate that cell-autonomous mechanisms govern how macrophages react to the same stimuli to differentially produce factors that facilitate regeneration.

eTOC

Macrophage secretory factors play distinct roles during tissue repair. Comparing tissue healing in spiny mice (regenerative) and laboratory mice (non-regenerative), Simkin and Aloysius et al., show that specific substances released by resident macrophages (e.g., VEGFC, LTF) facilitate complex tissue regeneration in spiny mice.

Graphical Abstract



Keywords

regeneration; macrophage; resident macrophage; spiny mouse; Lactotransferrin; Vegfc; lymphangiogenesis; wound healing; Csf1r

Introduction

Systemic depletion of macrophages in highly regenerative vertebrates (e.g., axolotls and zebrafish) interrupts regeneration^{1–3}. Similarly, neonatal heart, adult bone, and skeletal muscle regeneration are all inhibited by systemically depleting macrophages in mammals^{4–7}

and local depletion in adult mouse digits or spiny mouse ears pauses regeneration until macrophages infiltrate the healing tissue^{4–8}. As evidence accumulates supporting a pro-regenerative role for macrophages, their activity during the resolution phase of fibrotic repair (e.g. scar formation) supports a delicate balance between the time and tempo of macrophage infiltration and clearance as it relates to healing^{9–12}. For instance, timed macrophage depletion after injury reduces collagen production and improves functional recovery in liver, lung, and skin models^{13–17}, whereas delayed macrophage clearance is associated with collagen deposition in pulmonary fibrosis, cirrhosis and in keloid scars^{18–21}. Thus, while it remains unclear if macrophages specifically direct different healing outcomes across species, abundant evidence supports their necessary role in regenerative and fibrotic healing.

Despite a deep evolutionary history, macrophages have been best studied in mammals where their functional heterogeneity is attributed to developmental origin^{22–24}, tissue of residence^{25,26}, and the surrounding environment in which they exist^{27,28}. While an exhaustive number of *in vitro* experiments have highlighted how monocyte-derived macrophages transition to distinct phenotypic states in response to exogenous cytokines^{29,30}, more recent *in vivo* phenotyping supports a spectrum of phenotypes among infiltrating macrophages with overlapping transcriptional landscapes^{27,31–34}. Thus, within the same injured tissue at a single timepoint there exists a myriad of different macrophage subtypes that can shape, and be shaped by, the environment in distinct ways.

Although it is well known that macrophage variants can play contrasting roles during inflammation, the way in which these subtypes differentially coordinate the inflammatory phase of regeneration and fibrotic repair is less clear. In this regard, direct comparison of identical injuries with different healing outcomes can be informative. After cryoinjury to the heart, comparative studies using the zebrafish (regenerative) and medaka (scar-forming) found that macrophages enter the damaged tissue earlier and in greater numbers during regeneration³⁵ implicating inflammatory timing and magnitude as differences between regeneration and scar-formation. Parallel studies in the neonatal mouse heart (regenerative) and the adult mouse heart (scar-forming) suggest the predominant macrophage subtype (resident versus infiltrating) changes with age and that a shift towards a greater proportion of infiltrating macrophages contributes to a loss of regenerative ability^{15,35–37}. Together, these studies suggest that the tempo of infiltration and clearance, coupled with the phenotypic composition of the macrophage population, at least partially regulates the healing outcome.

Here we use a comparative system to study macrophage function during two healing outcomes in adult mammals: complex tissue regeneration and fibrotic repair. Spiny mice (*Acomys spp.*) can regenerate cartilage, hair follicles, skin, peripheral nerves, blood vessels and adipose tissue of the external ear pinna after a 4mm ear punch injury. An identical injury in the laboratory mouse (*Mus musculus*) results in scar formation^{38,39}. This study builds upon our previous work showing that *Acomys* have a robust macrophage response to injury but exhibit lower levels of key pro-inflammatory cytokines in the local microenvironment compared to *Mus* and other rodents^{40,41}. In this study, we identify two macrophage populations that persist through regeneration and scar-formation. We define intra- and interspecific phenotypic signatures among these populations and demonstrate a unique genetic signature for *Acomys* tissue resident macrophages prior to and during regeneration.

Moreover, we show *Acomys* macrophages secrete ligands (VEGFC, PDGFA, LTF) that promote a regenerative phenotype in fibroblasts from either species. Finally, we show VEGFC is specifically expressed by *Acomys* resident macrophages and functionally link its expression to blastema formation and regeneration. Overall, these experiments indicate that specific secretory factors of *Acomys* macrophages establish a tissue microenvironment conducive for regeneration.

Results

IFN γ +LPS stimulated bone marrow derived macrophages (BMDM) from *Acomys* possess a pro-regenerative secretome

Complete tissue excision from the skin or ear pinna provides a unique opportunity to track immune cell dynamics during regeneration in spiny mice (*Acomys spp.*) compared to fibrotic repair in outbred laboratory mice (*Mus musculus*)^{40–42}. Using a 4mm ear punch assay, we previously reported macrophage numbers during tissue healing in both species⁴¹. In this model, macrophages accumulate in the injured ear pinna, peak between D5–7 and are present through at least 20 days post injury⁴¹ (Figure S1A–C' and Figure S1D–E). However, using two common *Mus* macrophage markers (F4/80: pan-macrophage, CD206: anti-inflammatory/wound solving), we noted differences in the macrophage population structure. Whereas F4/80 and CD206 largely overlapped among *Mus* macrophages, they appeared to mark separate cell populations in *Acomys* at D5 (Figure S1F–G). These data raised several possible explanations for how macrophage populations could exhibit species-specific phenotypic compositions: either generic injury signals produce different macrophage phenotypes (cell autonomous regulation to similar injury response), or unique signals (via type or magnitude) differentially polarize macrophages to adopt cell states found during regeneration or fibrotic repair (extrinsic regulation of phenotype).

To determine whether naïve macrophages from *Acomys* and *Mus* differentially respond to the same stimulation signals, we tested the macrophage response to well-established *in vitro* stimulation paradigms used for mouse, humans, and rats^{43–47}. Since macrophage phenotype is influenced by tissue type and developmental origin^{18,24,27,31,48–50}, we confined our initial experiments to bone marrow derived macrophages (BMDM) (Figure 1A). BMDMs (M₀) were CD11b⁺;CD14⁺;IBA1⁺;CD68⁺ prior to *in vitro* polarization (Figure 1B). This data supported that BMDM from both species shared basic identity through a set of commonly used cell surface markers.

Previously we showed classical or alternative activation pathways could be induced in BMDM using IFN γ +LPS or IL-4, respectively⁴¹. Using qRT-PCR, we assessed the expression of established markers for these activation states in *Mus* and *Acomys* macrophages (Figure 1C). After IL-4 stimulation, *Cd206* expression significantly increased in *Mus* and trended upward in *Acomys* compared to IFN γ +LPS stimulation, while *Socs3* was significantly increased and *Tgfb1* was decreased in both species after IFN γ +LPS stimulation (Type III test of fixed effects, Table S1). Whereas *Cd86* expression in *Mus* was significantly upregulated with IFN γ +LPS treatment, *Acomys* macrophages did not express *Cd86* under either stimulation paradigm (Figure 1C, Table S1). These data suggested the same inflammatory cues could effectively polarize macrophages from both species, but also

showed stimulation could elicit subtle species-specific differences in macrophage phenotype. This led us to ask if secreted proteins from *in vitro* macrophage phenotypes would promote similar or alternate behavioral changes in primary ear pinna fibroblasts (Figure 1A).

To test this hypothesis, we isolated ear pinna fibroblasts from *Acomys* and *Mus* and cultured them for 48hrs in conditioned media from unstimulated (M_0), $M_{IFN\gamma+LPS}$ or M_{IL4} macrophages. To use ECM gene expression as one proxy for healing behavior in response to macrophage proteins, we assessed baseline expression of *Tenascin-C* (*Tnc*), *Fibronectin* (*Fn*), and *Collagen1a1* (*Col1a1*) in *Acomys* and *Mus* primary fibroblasts and found they were similarly expressed while *Matrix metalloproteinase 9* (*Mmp9*) was minimal in *Acomys* and *Mus* with comparatively lower expression in *Acomys* relative to housekeeping genes (Independent sample t-test, Table S2) (Figure 1D). While *Acomys* and *Mus* culture media from IL-4-stimulated macrophages (MCM- M_{IL4}) had little effect on *Tnc*, *Fn*, *Mmp9*, and *Col1a1* expression, $IFN\gamma+LPS$ -stimulated macrophage culture media (MCM- $M_{IFN\gamma+LPS}$) significantly altered fibroblast gene expression levels in a species-specific manner (Figure 1E, Figure S2).

We observed a significant decrease in *Col1a1* expression in response to *Mus* MCM- $M_{IFN\gamma+LPS}$ (*Mus* MCM on *Mus* fibroblasts) (Figure 1E, Table S3–4), although this result was shown to be $IFN\gamma$ and LPS-dependent, as fibroblasts responded similarly to control media alone (fresh media containing $IFN\gamma+LPS$) (Figure 1E). When *Acomys* fibroblasts were exposed to media from *Acomys* MCM- $M_{IFN\gamma+LPS}$ macrophages, *Col1a1* expression was significantly downregulated to a greater extent compared to *Mus*, indicating a stronger antifibrotic effect (Figure 1E, Table S3–4). *Acomys* fibroblasts significantly downregulated *Col1a1* when exposed to MCM- $M_{IFN\gamma+LPS}$ from either species but did not directly respond to $IFN\gamma$ or LPS in the control media. Additionally, *Acomys* MCM- $M_{IFN\gamma+LPS}$ significantly increased *Mmp9* expression in both *Acomys* and *Mus* fibroblasts, while *Mus* MCM- $M_{IFN\gamma+LPS}$ had no effect on *Mmp9* expression (Figure 1E, Table S3–4). Overall, these *in vitro* findings indicate that *Acomys* macrophages possess a distinct secretome compared to *Mus* when stimulated with $IFN\gamma$ and LPS, and their secreted products could promote a pro-regenerative ECM phenotype in fibroblasts from either species.

***Acomys* BMDM respond to pro-inflammatory cues by secreting low levels of pro-inflammatory cytokines**

Classically activated mouse macrophages ($M_{IFN\gamma+LPS}$) exhibit a pro-inflammatory phenotype and are known to secrete an array of cytokines^{51–53}. Given the differential response of fibroblasts to macrophage-secreted proteins from *Acomys*, we directly assayed cytokine levels in MCM (Figure 2A–C). Using a Quansys multiplex array validated for cross species analysis⁴⁰, we found that IL1a, IL12p70, TNFa, CCL2, CCL3, and CXCL1 increased following $IFN\gamma+LPS$ stimulation in both species (Figure 2A). However, CCL2, CXCL1, IL12p70, TNFa, IL5, IL6, and IL17 concentrations in *Mus* were significantly elevated compared to *Acomys* after $IFN\gamma+LPS$ stimulation (Figure 2A–B, Table S5). Only two cytokines from this panel, CCL3 and IL1a, were significantly more abundant in *Acomys* media from $M_{IFN\gamma+LPS}$ macrophages (Figure 2C, Table S5). Although these data support that *Acomys* $M_{IFN\gamma+LPS}$ macrophages are pro-inflammatory, our data indicate that *Mus*

BMDM adopt a significantly more pronounced inflammatory phenotype in response to IFN γ +LPS with the potential to propagate inflammation during wound healing.

Acomys* M_{IFN γ +LPS} macrophages specifically express the secreted factors *Pdgfa*, *Ltf*, and *Vegfc

We performed RNA-sequencing to identify potential secreted factors from *Acomys* M_{IFN γ +LPS} macrophages that might explain the observed regenerative fibroblast phenotype. We compared unstimulated (M₀) and M_{IFN γ +LPS} macrophages with an eye towards genes with little to no expression in M₀ macrophages and that significantly increased only in *Acomys* M_{IFN γ +LPS} macrophages. Among the 16,190 one-to-one orthologous genes analyzed, 624 were significantly upregulated in *Acomys* and *Mus* M_{IFN γ +LPS} macrophages, including known pro-inflammatory markers *IL1b*, *Cxcl11*, *Cxcl9*, *Cxcl10*, *Ccl8*, *Ccl2*, *Ccl7*, *Ccl12*, *Nod2*, and *Socs3* (Figure 2D – shared markers, Table S6)^{51,54–56}. Additionally, 847 genes were upregulated in *Acomys* and 594 in *Mus*, with some genes uniquely enriched in each species (e.g., genes associated with inflammatory macrophages *Cd22*, *Clec4e*, *Malt1*, *IL12b* were enriched in *Mus*, but not *Acomys* macrophages) (Figure 2D). Supporting our *in vitro* cytokine results, we found that *Il1a* was upregulated 13-fold in *Acomys* M_{IFN γ +LPS} and 7-fold in *Mus* M_{IFN γ +LPS} macrophages (Figure 2E).

Next, we isolated candidate genes that could explain our *in vitro* results which were (1) significantly upregulated in *Acomys* M_{IFN γ +LPS} macrophages compared to M₀ macrophages and (2) expressed in M_{IFN γ +LPS} but not in M₀ cells (Figure 2E). We also excluded genes that were expressed in *Mus* M_{IFN γ +LPS} and M₀ cells. Of 274 candidates (Table S7), we focused on 42 secreted proteins which included *Pdgfa*, *Lactotransferrin (Ltf)*, and *Vegfc*, known for their roles in fibrosis and regeneration^{57–59}. Notably, *Acomys* M_{IFN γ +LPS} macrophages expressed high levels of *Mmp9* and *Il10*, which are involved in matrix remodeling and anti-inflammatory responses (Figure 2E). Together, these data suggest that BMDM from both species exhibit distinct cell-autonomous responses to IFN γ and LPS stimulation, with *Acomys* macrophages uniquely secreting factors that promote angiogenic/lymphangiogenic (*Vegfc*), anti-inflammatory (*Il10*, *Ltf*), and matrix remodeling (*Mmp9*) activities that all contribute positively to regenerative healing.

PDGF-AA, IL1A, and LTF promote a regenerative phenotype in *Mus* fibroblasts

To extend our expression data, we asked if exogenous IL1A, LTF, PDGF-A, VEGFC or LTF, could promote a pro-regenerative phenotype. We treated *Mus* ear pinna fibroblasts with recombinant PDGF-AA, IL1A, VEGFC, or LTF and assayed for changes in *Col1a1* and *Mmp9* expression compared to treatment with *Acomys* M_{IFN γ +LPS} media (MCM) (Figure 2F–G). Exposure to PDGF-AA, IL1A, or LTF promoted a significant upregulation of *Mmp9* in *Mus* fibroblasts after 24 hours of stimulation whereas *Mmp9* expression did not change following VEGFC treatment (Two-Way ANOVA $p < 0.05$ with Dunnett's multiple comparison test, Table S8) (Figure 2F). In contrast, we did not see a significant change in *Col1a1* expression with IL1A, PDGF-AA or VEGFC (Figure 2G). These results support that PDGF-AA, IL1A, and LTF partially mimic the stimulatory effect of *Acomys* MCM leading to upregulation of *Mmp9*. Our observation that VEGFC did not alter *Mmp9* or

Col1a1 expression was not particularly surprising given the lack of VEGF receptors on *Mus* fibroblasts⁶⁰.

***Pdgfa*, *Ltf*, and *Vegfc* are expressed by *Acomys* resident macrophages during the inflammatory phase of regeneration**

Recent *in vivo* work has documented that macrophages in multiple healing paradigms do not adhere to the strict phenotypic categories recovered after various *in vitro* stimulations, although overlaps exist^{25,31,33,34,51}. To extend our *in vitro* experiments, we performed single-cell RNA-seq (scRNA-seq) on tissue harvested five days post injury (D5) during regeneration and fibrotic repair (Figure 3A and Methods). We sequenced approximately 4041 and 4325 cells from *Acomys* and *Mus* respectively. We identified twelve main clusters in *Acomys* based on the top differentially expressed genes and used established marker genes to classify five different keratinocyte populations, three fibroblast populations, a cluster of endothelial cells, T-cells and two macrophage populations (Figure 3A). From D5 tissue in *Mus*, we identified keratinocytes (six clusters), fibroblasts (two clusters), macrophages (three clusters) endothelial cells (two clusters), T-cells, chondrocytes and one cluster of unknown identity (Figure 3B).

In *Acomys*, we specifically identified two populations of macrophages. We identified one macrophage cluster as resident (*Csf1r*⁺;*Cd68*⁺;*Cd206*⁺)⁶¹ and the other as infiltrating (*Cd14*⁺;*Il1b*⁺;*S100a8*⁺) (Figure 3C,E, Table S9). Feature plot analysis revealed low levels of *Arginase 1* among all *Acomys* cells (Figure 3G). Like *Acomys* macrophages, we identified similar resident and infiltrating populations in *Mus* (Figure 3D–F). However, among the resident macrophages in *Mus*, we noted two separate clusters based on the expression or non-expression of the common alternative activation gene, *Arginase 1* (*Arg1*). Surprisingly, all resident *Acomys* macrophages were *Arg1*⁺ (*Acomys* and *Mus* comparison $p = 2.16 \times 10^{-21}$) (Figure 3H) and showed significantly lower levels of arginase activity (Two-way ANOVA main effects time and species, $F=53.4$, $p<0.0001$, $F=15.5$, $p<0.005$ respectively, * $p<0.0001$ Sidak's multiple comparison test between species at each timepoint) (Figure 3I).

We next asked if the four genes we identified in our BMDM data were expressed by macrophages *in vivo*. We found that infiltrating *Acomys* macrophages expressed *Ltf* at high levels. However, *Ltf* was the exception in this case. Instead, we found that it was the resident *Acomys* macrophage population which expressed all four secreted factors uncovered in our *in vitro* analysis: *Il1a*, *Pdgfa*, *Vegfc*, and *Ltf* (Figure 3J). Supporting the *in vitro* differences, we detected very low levels of these four genes in either *Mus* macrophage population (Figure 3K). Thus, our D5 *in vivo* scRNA-seq analysis established we could identify and compare macrophages from injured ear tissue across species, and highlighted that *Pdgfa*, *Vegfc*, and *Ltf* were expressed specifically by *Acomys* resident macrophages.

Two distinct populations of macrophages from *Acomys* and *Mus* maintain unique identities throughout regeneration and fibrotic repair

We extended our scRNA-seq analysis to include five timepoints: D0 (uninjured tissue), D3 and D5 (peak inflammation)⁴⁰ and D10 and D15 (new tissue formation or fibrosis) (Figure S3A–D). We first combined cells from both species across all timepoints and performed

data reduction by limiting our analysis to orthologous genes present in both species (approx. 16,190 genes). Following unbiased clustering, we found that clusters segregated by species (Figure S3E). To control for this strong species signal, we performed batch corrected dimension reduction using orthologous genes between species and excluded those that were unique to either species (Figure 4A). Again, unbiased clustering using genes with the highest variance revealed two populations of macrophages common across species; an *Il1b^{hi};Cd14^{hi};Csfr1^{low};Cd68^{low}* population (i.e., infiltrating) and a *Csfr1^{hi};Cd68^{hi};Cd206^{hi}* population (i.e., resident). Consistent with previous work⁶², we identified two universal macrophage markers from our dataset in the ear pinna, *Fcer1g* (the high affinity IgE receptor) and *Tyropb* (transmembrane immune signaling adaptor) (Figure 4A).

To identify macrophage phenotypes unique to each species and macrophage subtype, we ran unbiased clustering on the macrophage clusters from both species (Figure 4B). The main identifiers of each cluster were based on species first and macrophage subtype second (Figure 4B–C). By mapping time onto the cell clusters, we found that D0 and D15 cells almost exclusively clustered with resident macrophages (Figure 4C, black points D0, grey points D15) whereas D3, 5 and 10 were evenly distributed across both populations (Figure 4C). This finding is consistent with peripheral macrophages arriving after injury and then clearing/dying as inflammation is resolved.

To explore macrophage phenotypes, we used a recently established classification scheme for macrophages derived across multiple pathogenic models (i.e., inflammatory, phagocytic, resolution, and oxidative)³⁴ and mapped the top-regulated genes for each state onto our four macrophage clusters over time (Figure S4A). Although our clusters did not align with a single state from this scheme, we did observe *Acomys* resident macrophages were comparatively enriched for phagocytic and remodeling genes while resident and infiltrating macrophages from *Mus* exhibited an enrichment for the inflammatory signature (Figure S4A). Echoing our *in vitro* data, *Mus* infiltrating macrophages also exhibited an inflammatory and oxidative signature. Because our clusters did not fall within a specific activation profile, we used the top differentially expressed genes from our unbiased clustering to identify genetic markers to define each macrophage cluster. These markers included lysosomal proteases (i.e., *Ctss*, *Ctsz*) and immune regulatory membrane proteins in resident macrophages (i.e., *Emp3*, *Coro1a*) (Figure 4D, Table S10), and genes associated with infiltrating macrophages (e.g., *Srgn*, *Il1r2*, *Lcp1*, *S100a8*) upregulated by *SPP1^{hi};MERTK^{hi}* macrophages found in pulmonary fibrosis⁶³ (Figure 4D, Table S11). These genes provide markers for species-independent identification of infiltrating and resident macrophages in craniofacial tissue.

To isolate species-specific differences, we focused on genes uniquely expressed by each of the four clusters. *Mus* infiltrating macrophages expressed pro-inflammatory (*Trem1*, *Cd53*)^{64,65} and CD8 T-cell activation genes (*Mxd1*, *Cd52*)^{66,67}, none of which were highly expressed in *Acomys*. In contrast, *Acomys* infiltrating macrophages showed higher expression of genes involved in phagocytosis, bacterial clearance and TLR activation (*Ccl6*; *Crem*)^{68,69} and a serine protease inhibitor (*Slpi*) shown to reduce macrophage responses to LPS⁷⁰ (Figure 4E). Differentially expressed genes specific for resident macrophages in *Acomys* included the lectin recognition receptor *Ficolin B* (*Fcnb*), and *CXCL16*, an

membrane proteoglycan involved in phagocytosis of bacteria⁷¹. For *Mus* the common alternative activation markers, *Arginase1* and *Lgals3*, were highly expressed while not observed in *Acomys* resident cells. For the most part, these genes remained relatively consistent across time with a few exceptions that increased expression at Days 3–10 in the *Acomys* resident population. (Figure 4E). A GO analysis of differentially expressed genes across time revealed distinct pathway preferences, such as antioxidant and oxidoreductase pathways in *Acomys* and MHC activity in *Mus* resident macrophages (Figure S4B). After injury, resident macrophages in *Acomys* and *Mus* showed an increase in cytokine and actin binding pathways at D3–5. However, *Acomys* resident macrophages shifted toward terms associated with cell proliferation (mitogen-activated protein kinase binding, growth factor receptor binding) at D5 and antioxidant pathways at later time points (D10–15) (Figure S4B).

We next explored macrophage expression of the four secreted pro-regenerative candidates we identified from our *in vitro* study. Across all time points, *Ltf* was almost exclusively expressed in *Acomys* macrophages and mapped to the infiltrating and resident clusters (Figure 4F, Figure S4C). In *Acomys*, nearly all infiltrating macrophages expressed *Ltf* during healing (Figure 4F). Like *Ltf*, *Vegfc* expression was only detected in *Acomys* macrophages (Figure S4C). However, whereas infiltrating and resident macrophages were the source of *Ltf*, *Vegfc* expression was confined to resident macrophages (Figure 4F). *Pdgfa* expression was also confined to cells in the resident cluster, but we found that it was expressed in both species, albeit in a greater number of *Acomys* macrophages (Figure 4F, Figure S4C). These three genes were upregulated in *Acomys* resident macrophages as early as D3 after injury and remained highly expressed through D15 (Figure 4G). *Il1a* was more prevalent in the resident *Acomys* cluster at D5 compared to *Mus*, but at later timepoints, *Il1a* was more prevalent in the infiltrating *Mus* cluster compared to *Acomys* (Figure 4F, Figure S4C).

To validate and extend our single cell analysis, we used FACS to isolate and analyze macrophages from healing tissue at D5 using CD11b and CSF1R. FACS revealed two subpopulations of CD11b⁺ cells: a CSFR1⁺ and CSFR1[−] subpopulation present in the ear pinna (Figure 5A–B). Because *Csf1r* expression was consistently more prevalent in the resident population (compared to infiltrating macrophages), and because this population also expressed *Pdgfa*, *Ltf* and *Vegfc* (Figure S4C), we focused our analysis on CSFR1⁺ macrophages. CSFR1⁺ macrophages comprised 0.7% of the CD11b⁺ population in *Mus* compared to 36% in *Acomys* (Figure 5A, Figure S5). In both species, CSFR1⁺ cells expressed high levels of *Cd68* and lower levels of *Cd14* (Two-tailed Student's *t*-test, *Cd68*; *Acomys* *t*=3.84, *p*=0.018; *Mus* *t*=1.29, *p*=0.26) (Two-tailed Student's *t*-test, *Cd14*; *Acomys* *t*=0.57, *p*=0.59; *Mus* *t*=2.26, *p*=0.086) (Figure S5). CSFR1⁺ cells in *Acomys* expressed significantly higher levels of *Pdgfa*, *Il1a*, *Ltf*, and *Vegfc* and lower levels of *Arg1* compared to *Mus*; supporting our RNAseq datasets (Figure 5B, Two-tailed Student's *t*-test, *Ltf* *t*=9.44, *p*=0.0007; *Pdgfa* *t*=3.11, *p*=0.035; *Il1a* *t*=5.65, *p*=0.004; *Vegfc* *t*=3.6, *p*=0.022; *Arg* *t*=2.76, *p*=0.05).

To visualize the physical distribution of these gene transcripts among macrophage populations, we performed RNAscope on ear pinna tissue from D0 and D5. While we

identified some *Ltf*⁺ neutrophils (Figure 5C, yellow arrow)⁷², the majority of *Cd14*⁺*Ltf*⁺ cells showed nuclei characteristic of macrophages (Figure 5C, green arrow). By quantifying transcript number per cell, we observed that *Ltf* and *Vegfc* expression were upregulated at D5 compared to basal expression (*Ltf* Two-way ANOVA, main effect $F=172$, $p<0.0001$; *Vegfc* Two-way ANOVA, Main effect $F=158$, $p<0.0001$) (Figure 5C–F, Figure S6A–B). Both species showed *Pdgfa* induction at D5 with a higher prevalence in *Acomys Csf1r*⁺ macrophages (*Pdgfa* Two-way ANOVA, main effect $F=70$, $p<0.0001$) (Figure 5G–H, Figure S6C). Together, these data supported that the *Csf1r*⁺ macrophage population is the primary source of *Ltf*, *Pdgfa* and *Vegfc*. Finally, *Ltf* and *Vegfc* showed higher specificity to the *Acomys* regeneration phenotype, whereas *Pdgfa* was associated with general wound healing in both species (Figure 3J–K).

Macrophage secretory factors interact with fibroblast and endothelial cell receptors

To discern what cell types could respond to these macrophage-secreted ligands, we conducted receptor-ligand analysis using our scRNA-seq dataset during peak inflammation at D5 (Figure 6A–D, Table S12). This analysis revealed that a cluster of *Acomys* fibroblasts (Figure 3A) expressed *Pdgfrb*, supporting that *Pdgfa* expressed by resident macrophages could target these cells early during healing (Figure 6A). This interaction was not identified in the *Mus* macrophage and fibroblast clusters (Figure 6B). *Acomys* fibroblast populations also expressed receptors for *Vegfc* (*Itgb1*⁷³ and *Nrp2*⁷⁴) and *Ltf* (*Lrp1*^{75,76}) (Figure 6A–B). In addition, where *Ltf* and *Vegfc* were expressed from *Acomys* macrophages (Figure 6C, Table S12), endothelial cells expressed the corresponding ligands, *Itgbr* and *Kdr* for *Vegfc* and *Lrp1* for *Ltf* (Figure 6C). Completely different interactions were present in *Mus* at the same timepoint (Figure 6D). These results underscored the potential for macrophage-secreted ligands to affect stromal cell populations during the inflammatory stage of healing.

VEGFC protein is exclusively present during *Acomys* regeneration

To understand if the specific and elevated gene expression signature unique to *Acomys* macrophages was mirrored by elevated protein levels, we performed western blot analysis for LTF, VEGFC and PDGFA. Echoing our transcriptomic analyses, LTF and VEGFC were highly enriched during the inflammatory phase of regeneration in *Acomys* (D5 and D10) when macrophage numbers are high, after which protein levels returned to baseline (Figure 6E, Table S13). LTF, which was also enriched at D10 in *Mus*, showed a potential injury-induced modification in *Acomys* (~73kDa to ~78kDa shift) (Figure 6E, Table S13). VEGFC was exclusively observed in *Acomys* at D5 and D10. Examining PDGFA in *Acomys*, we found a smaller ~20kDa species enriched after injury whereas both a ~20kDa and ~28kDa species were present in uninjured *Mus* tissue, and weakly enriched after injury (Figure 6E, Table S13). While the molecular weight disparity in PDGFA across the two different healing outcomes needs further exploration, our protein data confirmed that VEGFC is a unique secretory protein product of *Acomys* macrophages during regeneration.

VEGFC is known for its role in lymph vessel formation during embryonic development and lymphangiogenesis in adults^{77,78}. Previous studies have implicated VEGFC-mediated lymphangiogenesis during tissue regeneration^{59,79,80}. Examining *Acomys* ear pinna, we found an extensive vascular and lymphatic network marked by smooth muscle actin (SMA)

and lymphatic vessel endothelial hyaluronan receptor 1 (LYVE1) respectively (Figure S7A) resembling that reported in *Mus* ear pinna⁸¹. To further explore VEGFC activity during *Acomys* regeneration, we analyzed receptor activation (Figure 6H). VEGFC-mediated signaling is initiated by the binding and phosphorylation of its receptors, VEGFR2 and VEGFR3, followed by the activation of a downstream signaling cascade in endothelial cells^{60,82}. In agreement with VEGFC ligand enrichment in response to injury, we also observed an increase in VEGFR3 phosphorylation (at tyrosine 1230) in *Acomys* at D10. Similarly, we observed an increase in VEGFR2 phosphorylation (at tyrosine 1175)⁸³ in *Acomys* at D10-D20 (Figure 6H, Table S14). In contrast, we did not observe activation of VEGFR2/3 in *Mus* during healing.

To further confirm that VEGFC is secreted by macrophages during *Acomys* regeneration, we depleted macrophages using clodronate-liposomes (see Methods)⁴¹. Macrophage depletion reduced both CSF1R⁺ macrophage numbers and *Vegfc* expression compared to control treated ears (PBS-liposomes), supporting that CSF1R⁺ macrophages secrete VEGFC during ear pinna regeneration in *Acomys* (Figure 6F–G). Taken together, our gene and protein data strongly suggest that the specific secretion of VEGFC from resident *Acomys* macrophages at least partially mediates VEGFR2 and VEGFR3 receptor activation during regeneration.

Inhibition of VEGFC signaling interrupts regeneration in *Acomys*

Given the concordance between VEGFC ligand and receptor activation in *Acomys*, we sought to functionally test the requirement for VEGFC-mediated signaling during regeneration. To inhibit the VEGFC ligand-receptor interaction, we used a blocking antibody to bind VEGFC. We used daily injections of the blocking antibody (Vegfc-BAb) or IgG control subcutaneously into the base of the ear from D0-D10 which corresponded to maximal VEGFC protein expression (Figure 6E). Phosphorylation of VEGFR3 was strongly inhibited at D10 and 20, while total VEGFR3 protein increased slightly in inhibited ears (Figure 7A, Table S15). Phosphorylation of VEGFR2 was reduced at D20, but not D10, suggesting compensatory interactions with other VEGFR2 ligands^{84,85} or reduced recruitment of endothelial cells (Figure 7A, Table S15). In comparison to the IgG control, blocking antibody treatment did not alter the either existing lymphatic or blood vessels in the uninjured area (Figure S7C–D). However, new tissue and hair follicle regeneration was reduced in BAb treated animals compared to IgG control (Two-tailed Student's *t*-test, $t=9.91$, $p=0.0005$) (Figure 7B–C). Similarly, there was a significant delay in closure rate from D15-D30 compared to IgG treated ears (Two-way ANOVA, main effect $F=99.11$, $p<0.0001$, Post Hoc analysis by Student's *t* pairwise comparison IgG vs VEGFC-BAb; D15 $t=-3.47$, $p=0.0007$; D20 $t=-3.56$, $p=0.0005$; D25 $t=-2.95$, $p=0.0038$; D30 $t=-2.16$, $p=0.0323$) (Figure 7D). Despite this early inhibition, treated animals ultimately recovered to regenerate their injuries (Figure 7D).

Because VEGFC-VEGFR3 signaling is known to direct lymphangiogenesis, we next analyzed lymphatic endothelial cells (LECs) using PROX1^{78,86,87}. We observed PROX1⁺ cells in the dorsal and ventral dermis on D10 in control animals (Figure 7E–F). Blocking

VEGFC significantly reduced total PROX1⁺ and actively proliferating (EdU⁺) PROX1⁺ cells (Two-tailed Student's *t*-test, D10 *t*=0.98, *p*=0.02; D20 *t*=.99 *p*=0.00004) (Figure 7E–K)

In addition to lymphangiogenesis, VEGFC can also stimulate angiogenesis via activation of VEGFR2 and VEGFR3 in endothelial cell^{88–90}. Blocking VEGFC reduced CD31⁺ cells and EdU⁺;CD31⁺ cells in the blastema at D20 (Two-tailed Student's *t*-test, EdU⁺, *t*=14.2, *p*=0.00014; CD31⁺ *t*=11.64, *p*=0.00031; CD31+EdU⁺ *t*=9.8, *p*=0.0006) (Figure 7L–O). Because some leukocytes can express CD31⁹¹ and VEGFC has a known role in inflammatory regulation^{92,93}, we stained tissue with CD31 and the pan macrophage marker IBA1. Indeed, a fraction of IBA1⁺ cells also stained positive for CD31, and treated animals exhibited a small, but significant increase in macrophage numbers suggesting prolonged inflammation after VEGFC blocking (Two-tailed Student's *t*-test, IBA1+CD31⁺ *t*=12.77, *p*=0.001; IBA1⁺ *t*= 4.94, *p*=0.015) (Figure 7P, Figure S7B). This data supports a previously reported anti-inflammatory role for VEGFC that may stem from reduced leukocyte drainage at the injury site^{92,93}. Taken together, our data supports that VEGFC secreted from *Acomys* resident macrophages plays a multifunctional role during regeneration by promoting lymphangiogenesis and angiogenesis and supporting inflammatory resolution and progenitor cell proliferation.

Discussion

Our results reveal several important insights into how macrophage populations behave in different rodent species and specifically how they regulate regeneration. We observed that IFN γ +LPS and IL4 can polarize macrophages from different murid rodents into pro-inflammatory and resolving phenotypes based on a small set of well-established markers identified for mouse and human⁹⁴. However, our data also showed that cell autonomous mechanisms trigger these same macrophages to produce species-specific factors. For instance, *Acomys* macrophages exhibited a muted inflammatory profile (compared to *Mus* and humans) by secreting low levels of certain pro-inflammatory cytokines as seen here and in other studies^{40,42}, while also secreting factors that antagonized collagen production and drove matrix remodeling in fibroblasts. Our *in vivo* experiments identified two primary macrophage populations that participate in tissue healing regardless of the healing outcome: infiltrating and resident. Importantly, we identified subtle interspecific differences between macrophage populations that aligned with regeneration compared to scar-formation. Specifically, we identified two secreted factors, LTF and VEGFC that were almost exclusively expressed by spiny mouse macrophages during regeneration, and other factors, PDGFA, that were upregulated by spiny mouse macrophages in response to inflammatory stimuli. Taken together, our findings revealed the importance of how subtle differences in macrophage phenotype could shift cells in the wound microenvironment towards regeneration or fibrotic repair.

Pro-regenerative versus fibrotic macrophages: is there a difference?

For the last decade, it has been unclear whether macrophage roles in regeneration overlap with those in scar-forming species^{95,96}. Studies have proposed common roles for macrophages in neutrophil and senescent cell clearance^{97,98}, but these roles are not

unique to regeneration¹⁴. Recent comparative models of regeneration and fibrotic repair have shown differences in the timing, magnitude, and cytokine profile of the inflammatory response^{35,37}. For example, in regenerative zebrafish hearts, more macrophages infiltrate faster compared to non-regenerative medaka hearts, and when macrophage infiltration into zebrafish hearts is delayed, neovascularization and regenerative ability are reduced³⁵. Although this study did not tease apart phenotypic similarities or differences in medaka and zebrafish macrophages, a study comparing regenerative and non-regenerative hearts following myocardial infarction from neonatal and adult mice respectively, found two different populations partially controlled the injury outcome^{15,99}. One population (CCR2⁻) in the heart was derived from the embryonic yolk-sac, was less inflammatory, and promoted coronary angiogenesis and cardiomyocyte proliferation to facilitate ventricular regeneration in neonates^{15,61,100}. In the adult heart, however, this population was replaced by infiltrating monocytes (CCR2⁺) after injury^{15,100}. The adult macrophages promoted inflammation and oxidative stress¹⁵ and directly contributed to collagen deposition³⁷. Supporting different functions among these populations, when neonatal macrophages were transplanted back into older mice, better functional recovery occurred¹⁰¹. These studies support that while the timing and magnitude of the inflammatory response is critical to determining healing outcome, phenotypic differences can significantly alter how macrophages impact nearby cells. Importantly, results from these studies also suggest that cell lineage influences how macrophages regulate inflammatory magnitude and resolution.

Macrophage infiltration has been well-defined for scar-forming injuries^{9,102}, but the origin of these macrophages is still unknown¹⁰³. Cells of the mononuclear phagocytic system that reside in steady state epidermis and dermis are a heterogeneous population of macrophages and dendritic cells with different developmental origins^{24,25,104–107}. In many tissues, adult macrophages self-replenish during normal tissue turnover and after inflammation where they do not require a contribution from bone marrow or circulating monocytes¹⁰⁸. In the skin, and in musculoskeletal tissues the lineage of different macrophage subsets during homeostasis prior to tissue injury is unclear^{24,109}. Using scRNA-seq to analyze all time points together, we reliably detected two macrophage populations in both species: a population that was primarily *Cd14⁺;Il1b⁺;S100a8⁺;Cd68⁺;Csf1r⁻;Cd206⁻* (for simplicity the *Cd14⁺* population), and a population that was primarily *Cd14⁺;Il1b⁻;S100a8⁺;Cd68⁺;Csf1r⁺;Cd206⁺* (for simplicity the *Csf1r⁺* population). Although we did not perform cell-tracing experiments, we could nevertheless detect the *Csf1r⁺* population in uninjured tissue by analyzing our scRNA-seq and RNAscope data. In contrast, the *Cd14⁺* population was sparse prior to injury, whereby it significantly increased five days later, a finding that echoes infiltration kinetics of circulating macrophages following skin injury. This suggests that the *Csf1r⁺* population we found co-expressing *Ltf*, *Vegfc*, and *Pdgfa* in *Acomys* is a resident population which expands upon injury.

Other studies utilizing scRNA-seq have highlighted distinct macrophage populations under fibrotic conditions^{110–113}. Although consensus gene markers for a fibrotic macrophage found across tissues has remained elusive, we observed some previously identified fibrotic markers in *Mus* macrophages during repair of the ear. Interestingly, several of these “fibrotic” markers were also present in *Acomys* macrophages during regeneration. This

suggests that (1) these genes are broad markers of a wound healing macrophage, (2) fibrotic macrophages present in *Acomys* ears are outnumbered by other macrophages present in the wound microenvironment, or (3) surrounding cells in *Acomys* ears do not respond to the fibrotic macrophages in the same way. Further comparative studies will pinpoint where the overlap and differences exist for macrophage subtypes, markers, and roles during regeneration and fibrotic healing.

Regenerative macrophages stimulate extracellular matrix turnover in *Acomys*

Extracellular matrix turnover is associated with high MMP activity and is a conserved feature of vertebrate regeneration^{38,114–120}. In response to injury in urodele limbs and skin, *Mmps* are upregulated early in epidermal cells and later in blastemal cells^{115,118,120–123}. Lower expression of *Mmp9* and higher collagen deposition is correlated with regenerative failure in *Xenopus* froglets, the short-toe mutant axolotls¹²⁴ and in denervated axolotl limbs¹²⁰, while higher expression of *Mmp9* is associated with scar-free skin healing in young, athymic mice¹²⁵. This evidence across regenerating models, while correlative, suggests histolysis may be necessary to free cells from surrounding extracellular matrix (ECM) for regeneration-competent cells to proliferate and participate in regeneration. Alternatively, active matrix degradation may simply limit excess collagen accumulation that is a prominent feature of fibrotic repair. Our results provide cellular context to previous work on whole tissue by showing that *Mmp* expression can be directly influenced by macrophage-secreted signals. Specifically, we identified three factors (PDGFA, IL1A, LTF) which were uniquely secreted by *Acomys* macrophages and that could induce *Mmp9* expression in *Mus* fibroblasts. Interestingly, PDGFA can be pro-fibrotic depending on the tissue context as has been demonstrated in the lung¹¹⁰. Our findings suggest that within the context of epimorphic regeneration, PDGFA might instead be more in line with studies in digit regeneration models, where PDGFRa is a blastemal cell marker⁵⁷. Therefore, PDGFA secreted by *Acomys* macrophages may help in recruitment of progenitor cells as well as support matrix turnover by ear fibroblasts.

Acomys macrophages exhibit a dampened inflammatory profile compared to *Mus*

Our *in vitro* finding of reduced inflammatory cytokines in *Acomys* macrophages parallels *in vivo* findings in the injury microenvironment conducted across wild and lab populations of *Acomys* and *Mus*⁴⁰. Previous studies in *Acomys* suggested lower levels of inflammation are conserved across injury sites from kidney to skin^{42,126,127} and that each tissue also shows better functional recovery after injury when compared to mouse. While these previous studies suggest lower inflammatory cell contribution drives lower inflammatory levels, our data suggests that a reduced inflammatory response is due to an intrinsic difference in the inflammatory cells themselves. Whereas *Mus* produce a strong cellular response to bacterial challenge, *Acomys* have unique bacterial killing properties that primarily involve non-cellular blood components (e.g., complement)¹²⁸. It is possible this immune difference may cross over to differences in response to injury as well. Recent studies in the axolotl have shown axolotl macrophages display prolonged ERK and MAPK signaling in response to damage associated molecular patterns plus LPS (and other PAMPs). On the other hand, mouse macrophages downregulate ERK and MAPK signaling in response to the same stimulation¹²⁹. The fact that axolotl macrophages respond to injury and infection

through upregulation of proliferative pathways instead of pro-inflammatory pathways may be a common thread for regenerative systems. Of note, LTF has been previously shown to have anti-senescent effects by reducing expression of monocyte-derived inflammatory cytokines^{130,131}. We found increased *Ltf* in bone marrow derived macrophages stimulated with LPS and IFN γ suggesting that a reduced inflammatory environment *in vivo* parallels an increase in LTF secreted from macrophages.

VEGFC signaling supports blastema formation and regeneration in *Acomys*

Macrophages are critical for neovascularization and lymphangiogenesis during inflammation^{132,133} and both processes support regeneration in a context specific manner^{59,79}. Moreover, resident macrophages play an important role in lymphatic vessel development and patterning¹³⁴. Our study provides multiple lines of evidence (RNA and protein) documenting that VEGFC is secreted from *Acomys* resident macrophages and functional data showing that VEGFC is essential for lymphangiogenesis and neovascularization in the regenerating *Acomys* ear pinna tissue. Importantly, VEGFC was completely absent during fibrotic repair in *Mus*. VEGFC has been well characterized in embryonic and adult lymphangiogenesis via activation of VEGFR3^{60,135} and VEGFC can stimulate an angiogenic response among endothelial cells^{88–90}. Growing evidence suggests that the lymphangiogenic response to injury is important for complex tissue regeneration. For example, induction of *Vegfc* and the proteolytic activation of VEGFC protein are crucial in zebrafish tail fin regeneration¹³⁶ and *Vegfc* or *Vegfr3* zebrafish mutants exhibit impaired lymphangiogenesis and regeneration⁷⁹. Similarly, VEGFC is important for normal cardiac repair in mice. In injured mouse hearts, macrophages produce VEGFC in response to myocardial infarction. Inhibiting VEGFC reduces lymphangiogenesis while increasing inflammation, thereby inhibiting a cardioprotective role for macrophage-secreted VEGFC in response to myocardial infarction¹³⁷. Similarly, we found that blocking the VEGFC-receptor interaction significantly impaired new tissue formation during *Acomys* ear pinna regeneration. This was due, in part, to a significant reduction in LECs (PROX1⁺) and endothelial (CD31⁺) cells in healing tissue.

During tissue healing, the lymphatic network contributes to inflammatory resolution by serving as a drainage system for macromolecules and immune cells and retarded lymphangiogenesis leads to increased fibrosis through reduced immune cell clearance; a result which promotes persistent and chronic inflammation⁹³. Similarly, we observed elevated macrophage numbers after VEGFC inhibition, although it remains unresolved whether this resulted from deficient clearance or enhanced recruitment of circulating macrophages. In parallel with delayed blastema formation, hair follicle neogenesis was inhibited following VEGFC blocking, in agreement with previous reports implicating VEGFC and lymphangiogenesis in hair follicle formation and growth^{138,139}. Interestingly, we also observed PROX1⁺ cells in the follicular epithelium of newly developing hair follicles (but not old ones) during regeneration. The identity of these epidermal specific PROX1⁺ cells have not been previously reported and further studies are required to resolve their role, if any, during hair follicle neogenesis. PROX1 is well known as master regulator of LEC fate, and it has also been studied in cancer stem cells^{140–142}. It is intriguing to consider that PROX1 may label a broader progenitor cell pool beyond LECs.

Ultimately, our data supports that resident macrophages help establish conditions in the injury microenvironment in regenerative *and* non-regenerative tissues while highlighting subtle phenotypic differences across species that appear to direct healing along two separate trajectories. Based on our data, we propose a working model during spiny mouse regeneration where (i) resident macrophages maintain a sterile wound through the use of lectin receptors and enzymes for bacterial cell breakdown, (ii) secrete factors like LTF to reduce inflammatory magnitude and limit oxidative stress in nearby stromal cells, (iii) secrete VEGFC to stimulate rapid angiogenesis and lymphangiogenesis, and (iv) secrete additional factors that promote MMP production and matrix remodeling to antagonize collagen production. Together these actions create a regeneration-permissive environment not observed in mouse.

Limitations of the study

While our scRNA-seq approach provided valuable information about macrophage heterogeneity, shallow sequencing depth associated with this methodology means we may have failed to capture smaller, less abundant macrophage subpopulations. Additionally, focusing on genes annotated in both species means we may have ignored unique species-specific gene combinations that might contribute to regeneration in *Acomys*. Future studies integrating transcriptomic and genomic analyses with larger sample sizes will provide a complete atlas of macrophage diversity during regeneration and fibrotic repair. Additionally, while our study highlighted the importance of select macrophage-secreted factors in ear pinna regeneration, it primarily focused on the functional requirement for Vegfc during the inflammatory phase of regeneration. Extended inhibition of Vegfc-signaling was not conducted leaving questions about the long term impact of impaired lymphangiogenesis on ear pinna regeneration unanswered. A logical next step would be to examine the effects of Vegfc inhibition in combination with other macrophage-secreted factors. To gain a comprehensive understanding of the mechanisms involved in ear pinna regeneration, future research should explore potential compensatory pathways that may be activated in response to Vegfc inhibition. These compensatory mechanisms could reveal additional layers of complexity in the regulation of macrophage activation states and tissue repair. Finally, our study primarily addressed the impact of Vegfc on macrophage behavior without distinguishing between its vasculogenic-dependent and -independent functions. Investigating these distinct roles of Vegfc in ear pinna regeneration could provide more nuanced insights into its contributions to the healing process.

STAR Methods

RESOURCE AVAILABILITY

Lead Contact—Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Ashley W. Seifert (awseifert@uky.edu).

Materials Availability—RNAscope®Probe are available through [acdbio.com](https://www.acdbio.com). Beyond these probes, this study did not generate new unique reagents.

Data and Code Availability—The scRNA-seq data are available through the Gene Expression Omnibus (GEO) under the accession number GSE182141. Any additional information not contained in Supplemental Information that is required to reanalyze data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animal care and Ear punch assay—*Acomys cahirinus* and *Mus musculus* (Swiss Webster, Envigo Hsd:ND4) were housed at the University of Kentucky, Lexington, KY. *A. cahirinus* were housed at a density of 10–15 individuals in metal wire cages (24 in. × 18 in. × 16 in., height × width × depth) (Quality Cage Company, Portland, OR) or in standard rat cages at a density of 3–5 individuals and fed a 3:1 mixture by volume of 14% protein mouse chow (Teklad Global 2014, Harlan Laboratories, Indianapolis, IN) 1x/day and black-oil sunflower seeds (Pennington Seed Inc., Madison, GA) 2x/week¹⁴³. *Mus* were fed mouse chow only. *Acomys* and *Mus* were exposed to natural light, and all animals used were sexually mature. Experiments used a combination of male and female animals matched between species. Animal age group used for the study: *Mus*, 3–6 months and for *Acomys*, 6–12 months. For ear punch, animals were anesthetized with 3% vaporized isoflurane (v/v) (Henry Schein Animal Health, Dublin, OH) at 1 psi oxygen flow rate. Macrophage depletion by Clodronate-Liposome or the control PBS-Liposome injection performed as previously reported¹⁴⁴. A 4mm biopsy punch (Sklar Instruments, West Chester, PA) was used to create a through-and-through hole in the right and left ear pinna. Ear tissue was collected at specified time points with an 8mm biopsy punch (Sklar Instruments, West Chester, PA) circumscribing the original injury. All animal procedures were approved by the University of Kentucky Institutional Animal Care and Use Committee (IACUC) under protocol 2019–3254.

Bone Marrow Derived Macrophage (BMDM) analysis—Macrophage progenitors were isolated from femur and tibia of *Acomys* and *Mus* as previously described⁴¹. Briefly, after sacrifice, the femur and tibia were surgically removed, and mechanically cleared of all skin, muscle, and tendon. Marrow was aspirated from bones by flushing the marrow with 10 ml of RPMI + 10% FBS through a 28-gauge syringe. Red blood cells were lysed with a hypotonic solution and remaining cells were plated at a density of 1×10^6 cells/ml in T-75 culture flasks. For the first seven days, bone marrow cells were grown in complete RPMI 1640 media (Gibco) supplemented with 20% L929 media containing M-CSF⁴⁴, 10% heat-inactivated fetal bovine serum (FBS) (Gibco), and 1% penicillin/streptomycin (PenStrep). After seven days in culture, BMDM were split using cold PBS and a cell scraper, plated onto coverslips in 24 well plates at a density of 5×10^5 cells/ml and allowed to settle for 24 hours in complete RPMI 1640 media supplemented with 10% FBS and 1% PenStrep. For macrophage activation, cells were stimulated with either 500 μ l of IFN γ (20 pg/ml) and LPS (*E. coli* derived, Sigma L3012, 200ng/mL) in RPMI media with 10% FBS or 500 μ l of IL-4 (20ng/ml) in RPMI media with 10% FBS. 24 hours after activation, media was collected and filtered to remove floating cells. Media was divided for cytokine analysis and fibroblast stimulation. Cells were collected in Trizol for RNA isolation.

Fibroblast activation assay—Ear fibroblasts were isolated from uninjured *Mus* and *Acomys* ears as previously described¹⁴⁶. A 4 mm ear punch biopsy was used to remove a full-thickness piece through the center of the ear pinna. Tissue was pooled from both ears of a single individual and single cell suspensions were made by enzymatic (Trypsin/Dispase) and mechanical digestion (chopping with blade/passing through 70 μ m filter). Cells were cultured in complete-DMEM media (Gibco) with 10% FBS (Hyclone) and 1% Antibiotic-Antimycotic (Gibco) and passaged once. Fibroblasts were cultured at 3% oxygen in 24 well plates. At 70% confluency, cells were treated with either Macrophage Conditioned Media (MCM) (from macrophages stimulated with either IFN γ +LPS or IL4) at concentrations of 100%, 50% and 5% diluted in DMEM, recombinant human Lactotransferrin (Lifespan Biosciences LS-G145517, 180604) at 3000, 300, 30 ng/ml in DMEM, recombinant human IL1a (Peprotech 200-01A) at 300, 30, 3 ng/ml in DMEM, recombinant mouse VEGFC (BioLegend 775104) at 500, 50, 5 ng/mL in DMEM, or recombinant mouse PDGF-AA (Peprotech 100-13A) at 10, 1ng/ml in DMEM. Each assay contained n=3 technical replicates. After 24 hours, fibroblasts were harvested in Trizol for RNA isolation.

METHOD DETAILS

VEGFC-Blocking experiment—Immediately following a 4mm ear punch, rabbit VEGFC-C blocking antibody (1 μ g/10 μ l, Cat#: pV1006R-r, ANGIO-PROTEOMIE) or rabbit IgG control (1 μ g/10 μ l, Cat. #: I-1000, Vector Laboratories) was injected via Hamilton syringe at the base of the injured ear once daily and continuing through D9. EdU (10 μ g/g body weight) was injected once every three hours for a total of three injections with tissues collected three hours after the final injection. Animals were euthanized and perfused with PBS to reduce autofluorescence from red blood cells. Collected tissues were used for immunohistochemistry and protein isolation. Ear hole closure measurement was performed following previously reported protocols³⁸.

Tissue preparation—Immunofluorescent staining was performed on formalin fixed, paraffin embedded tissue (FFPE) except for PROX1 (cryosections) or in whole mount for LYVE1 and SMA. Harvested tissue was placed into 10% (v/v) neutral buffered formalin (NBF, American Master Tech Scientific Inc., Lodi, CA) and incubated 9hrs for cryosections or 16–18hrs for FFPE at 4°C. For cryosections, D10 and D20 fixed tissues were cryoprotected in 30% sucrose until equilibration, then divided along the proximal-distal axis and embedded in Tissue-Tek OCT compound (Cat. #: 4583, Sakura) in an isopentane bath using liquid nitrogen. Tissues were cut at 10 μ m using a Thermo Cryostat NX50 cryostat, put on gelatin subbed slides, placed on a slide warmer overnight at 45°C, and then stored at –20°C until immunostaining.

For paraffin embedded samples, fixed tissue was washed three times with PBS, three times with 70% (v/v) ethanol and stored in 70% (v/v) ethanol at 4°C. All FFPE tissue processing was completed using a rapid microwave histoprocessor (Micron Instruments, Inc. Carlsbad, CA). Tissues were embedded in paraffin (Cat. #: 38–1450, Leica Biosystems, Buffalo Grove, IL) and 5 μ m sections were placed onto Superfrost Plus slides (Fisher Scientific). Tissue was subsequently deparaffinized in xylene and rehydrated through a series of graded

EtOH washes. For histology analysis tissue samples were stained with Masson's trichrome (Cat. #: KTMTRLT, StatLab) as per manufacturers instruction.

Whole mount tissues were harvested under isoflurane anesthesia, then post-fixed in 10% NBF for one hour. The dorsal epidermis was carefully removed, and ears were returned to NBF for an additional 4 hours then washed 6x in PBS before staining.

Immunohistochemistry—FFPE tissue sections underwent heat-induced epitope retrieval using buffered Tris-EDTA pH 9.0 in a steamer at 90°C for 15 minutes. Cryosections were allowed to return to room temperature. EdU labelling was according to manufacturer's protocol (Invitrogen). Briefly samples were incubated 30mins in the dark with EdU cocktail containing 5µl of 2M Tris (pH 8.5), 2µl of 50mM CuSO₄, 0.2µl of Alexa Fluor[®] azide 594, and 20µl of 0.5M ascorbic acid dissolved in water for a final volume of 100µl. Samples were blocked using Donkey/Goat serum for 30 min. All tissues were incubated with primary antibodies for 36 hours (whole mount) or overnight: rat anti-mouse F4/80 (1:100, Cat. #: 565409, BD Biosciences), IBA1 (1:1000, Cat. #: NB100–1028, Novus), CD31 (1:200, Cat. #: AF3628, R&D systems), LYVE1 (1:100, Invitrogen Cat. #: 13044382), smooth muscle actin (SMA, 1:100, Cat. #: NB300–978, Novus) or PROX1 (1:100, Cat. #: 11–002P, AngioBio). Secondary detection of antibodies was carried out for 30 minutes or 2 hours (whole mount) using streptavidin or donkey antibodies conjugated to Alexa Fluors (1:500, Invitrogen, Carlsbad, CA). For all sections, nuclei were counterstained with 10 µg/ml Hoechst 33342 (Cat. #: H3570, Invitrogen, Carlsbad, CA) and coverslips were mounted using ProLong Gold mounting medium (Invitrogen, Carlsbad, CA). Photomicrographs within the center of the injury were obtained at 4x (whole mount) or 20x magnification using an Olympus IX83 or BX53 fluorescent deconvolution microscope (Olympus America Inc).

Protein isolation and Western blotting—Healing ear tissue was collected, snap frozen in liquid nitrogen, and samples were stored at –80°C until processing. Samples were thawed in RIPA buffer (ThermoFisher) with protease inhibitors (cOmplete[™], Mini, EDTA-free Protease Inhibitor Cocktail, Cat. #: 11836170001, MilliporeSigma) and phosphatase inhibitors (Simple Stop[™] 2 Phosphatase Inhibitor Cocktail, Cat. #: GB-451-1, GOLDBIO) and the samples homogenized using a pellet pestle. Samples were sonicated for 30sec, 3x on ice with 30sec intervals. The lysate was centrifuged at 13,000 rpm for 15 mins at 4°C and the supernatant was collected and stored at –80°C. Protein concentrations were quantified via BCA assay kit (Pierce, Thermo fisher scientific).

Western blot samples were prepared using 25ug protein with 4X LDS sample buffer and 10X reducing agent. Samples were denatured at 70°C for 10min. Samples were run on 15-well, 4–12% Bis-Tris gels (NuPAGE[™]-NP0336) with MES SDS running buffer supplemented with 500µl of NuPAGE antioxidant added to the chamber. Electrophoresis was done at 180V for 45 mins and gels were transferred to PVDF membranes using a wet transfer Bio-Rad apparatus, with a constant current of 400mA for 5 mins on ice. Blots were washed with 1X TBST (TBS + 0.1% Tween-20) buffer. Membranes were blocked with 1% fish gelatin (for phospho-antibodies) or 5% milk (for other antibodies) for 1h. Primary antibodies were diluted in blocking buffer and incubated over night at

4°C. Primary antibodies used: Lactoferrin (LTF) (1:1000, Cat. #: PA5-95513, Invitrogen), VEGFC (1:1000, Cat. #: NB110-61022, Novus Biologicals), PDGFA (1:1000, Cat. #: NBP2-99113, Novus Biologicals), Phospho-VEGFR3 (1:1000, Cat. #: orb1095455, biorbyt), VEGFR3 (1:1000, Cat. #: orb315557, Biorbyt), Phospho-VEGFR2 (1:1000, Cat. #: 2478, Cell Signaling Technology), VEGFR2 (1:1000, Cat. #: 2479, Cell Signaling Technology), GAPDH (1:2000, Cat. #: 2118, Cell Signaling Technology). Membranes were washed 3x with 1X TBST for 5 mins each. Anti rabbit-HRP (1:2000, Cat #sc-2030, Santa Cruz Biotechnology) was used as a secondary antibody, diluted in blocking buffer, incubated with the membrane for 30 mins at room temperature and washed 3x with 1X TBST. Protein bands were detected with the Clarity™ ECL detection kit (BIO-RAD, Cat. #170 –5060) as per manufacturers protocol, and images were captured using a chemiluminescence imager. Western blots were quantified by image intensity analysis using FIJI software¹⁴⁵.

Flow cytometry analysis—M0 macrophages were fixed in 2% formaldehyde for 15 min at room temperature. Cells were permeabilized with 0.1% Triton-X100 in PBS for 10 min at room temperature. Cells were blocked with 5% goat serum. Cells were incubated in primary antibody for 1hr at room temperature. Primary antibodies used: CD14 (1:100, Cat. #: 17000-1-AP, Proteintech), CD68 (1:100, Cat. #:ab955, Abcam), CD11b (1:200, Cat.#:NB110-89474, Novus Biologicals), IBA1 (1:100, Cat.#:NB100-1028), Rabbit IgG (1:100, Cat.#:I-1000, Vector Laboratories), Mouse IgG (1:100, Cat.#:I-2000, Vector Laboratories). Secondary detection of antibodies was carried out for 30 minutes using goat antibodies conjugated to Alexa Fluors (1:500, Invitrogen, Carlsbad, CA). Cells were detected using BD FACSymphony™ A3 Cell Analyzer (BD Biosciences, USA) and data analyzed with FlowJo software (version 10).

Multiplex cytokine analysis—Macrophage conditioned media was assayed in duplicate using a custom-designed, multiplexed, sandwich ELISA array (Quansys Biosciences, Logan UT). The custom assay was designed to measure 16 antigens including IL-1a, IL-1b, IL-2, IL-4, IL-5, IL-6, IL-10, IL-12p70, IL-17, CCL2, CCL3, CCL5, CSF2, IFNγ, TNFa, and CXCL1. The assay was performed as previously described⁴⁰. Briefly, media was collected from each well of a 24 well plate. The primary antibody cocktail was loaded, and the plate was incubated at 4°C for 8 h on a plate shaker set to 500 rpm to allow binding of the biotinylated detection antibodies to the captured antigens. After washing the plate 4x, streptavidin-HRP conjugated secondary antibody cocktail was loaded and the plate was incubated at room temperature for 30 min on a plate shaker set to 500 rpm. The plate was washed 8x, chemiluminescent reagent was added, and the plate was immediately imaged with a chemiluminescent plate imager set to the manufacturer recommended image capture settings (Q-view imager, Quansys Biosciences). Our previous parallelism analysis of IL10 and RANTES cross-species specificity suggest these antibodies do not cross-react well with *Acomys* proteins, thus we excluded these two proteins from our final analysis⁴⁰.

RNA-seq analysis—Macrophages from *Acomys* (n=4) and *Mus* (n=4) were isolated and stimulated with IFNγ+LPS as described above. RNA was isolated from stimulated and unstimulated cells, frozen, and sent for sequencing at the Genomic Services Lab at Hudson Alpha. An indexed sequencing library was created from total RNA using the PolyA method

with Ribosomal Reduction. RNA was analyzed on a HiSeq v4 PE50 at 250 mil reads per lane. Sequencing of 8 samples across one lane yielded approximately 21 million reads per sample. Quality scores (QScore) were >36 for all samples. Tophat2 was used to align raw reads from each species to their respective genomes¹⁴⁷. For *Acomys*, the genome and annotation generated by the Rodent Genomes Consortium (paper in preparation, available for rapid release through ENSEMBL) was used for this analysis ([GCA_907164435.1 \(mAcoDim1_REL_1905\)](https://www.ensembl.org/Acomys)). For *Mus*, the mm10 genome (<https://hgdownload.soe.ucsc.edu/downloads.html#mouse>) assembly and annotation was used. After alignment, Cufflinks¹⁴⁸ was used to generate estimates of gene expression (FKPM) for each species and condition and Cuffdiff was used to determine gene expression fold changes between stimulated vs. unstimulated macrophages in each species independently¹⁴⁹. Expression level was set at > or < 20 FPKM to quantify a gene as expressed versus not expressed.

GO Analysis—Genes enriched per timepoint in the four macrophage types, the two species' infiltrating and resident macrophages, were determined using Seurat's find markers function. GO terms for the macrophages were determined using the enrichGO function in the R library clusterProfiler^{150–153}. The top 5 GO terms ranked by qvalue were plotted on the GO term heatmap.

FACs analysis—An 8mm ring of tissue circumscribing the injury area was collected post injury at D0 and 5. To create a single-cell suspension, we used enzymatic and mechanical digestion. Tissue was digested with a 1:1 0.25% Trypsin-EDTA (ThermoFisher, Cat #25200056)/Dispase (Corning, Cat #354235) solution for 1 hour at 37°C allowing for subsequent mechanical separation of the epidermis. Separated epidermis and dermis were incubated with a solution of 0.04g/ml collagenase Type 1 (Gibco, Cat #17100017), in HBSS (VWR 45000–456) for 1 hour at 37°C. Following digestion, cell suspensions were washed with PBS and filtered through a 70 µm cell strainer. Single cell suspensions were incubated 15 mins with an FcγR block (CD16/32 block, 20 µg/ml, BioLegend, Cat #101302) followed by incubation with directly conjugated primary antibodies at 4°C for 1 hour. Antibodies included PE-conjugated CD11b (BD Pharmingen, Cat #557397, 3µg/mL) and APC-conjugated CSF1r (CD115) (BioLegend, Cat #347306) diluted in FBS-staining buffer (BD Pharmingen, Cat #554656). Fluorescent activated cell sorting (FACS) was carried out by trained experts in the University of Kentucky Flow Cytometry Core using the iCyt Synergy sorter system (Sony Biotechnology Inc., San Jose, CA). Laser calibration was performed for each experiment using unstained, single fluorescent, and fluorescent minus one (FMO) control samples. Dot plots were created using FloJo (version 10). Cells were sorted to RNeasy Lysis Solution (Qiagen Cat # AM7020) for RNA analysis.

Quantitative PCR—For RNA analysis, cells were collected in 400µL Trizol reagent. RNA was isolated using Trizol: Chloroform extraction (10:1) with isopropanol precipitation (1:1 isopropanol to Trizol) and two 70% EtOH washes before drying the pellet and resuspending the RNA in water. RNA isolation from FACS sorted cells done by RNeasy Plus Micro Kit (Qiagen Cat #74034) as per manufacturers protocol. RNA quantity and quality was checked on a Nanodrop optical density reader. 200–500 ng of RNA was then converted to cDNA

using SensiFAST cDNA synthesis Kit (Thomas Scientific) according to the manufacturer instructions. Quantitative PCR was carried out using SYBR Green (QuantaBio) on a Roche 96 light cycler with species specific primers (Table S16).

Single cell analysis (scRNA-seq)—Tissue was collected from injured *Mus* and *Acomys* ears at D0, 3, 5, 10, and 15 after injury. An 8mm biopsy punch was used to collect tissue circumscribing the injury. Tissue was collected from both left and right ears from one animal per timepoint (23mg tissue). Tissue was placed in cold Hypothermosol FRS (Sigma, H4416) after collection then transferred to a petri dish on ice and minced with a razor blade into 1mm³ pieces. Minced tissue was incubated with 10mg/mL protease from *Bacillus Licheniformis* (Sigma, P5380) in DPBS containing 0.5mM EDTA, according to published protocol¹⁵⁴. Tissue was digested on ice for 20 minutes with vigorous mixing and trituration. Digested tissue was then transferred to a dounce homogenizer (Bellco, 1984–10002) and ground for 30 minutes in enzyme mix. Supernatant was filtered through a 70µm cell strainer. Leftover tissue was ground in fresh enzyme mix for an additional 15 minutes and then added to the 70µm filter to create a single cell suspension. Red blood cells were removed with RBC lysis buffer (Sigma, R7757). Cell count and viability was analyzed with trypan blue under an inverted microscope. Cell yield was on average 9000 cells/mg with 98% viability.

The single suspension was loaded into a well on a 10X Chromium Single Cell instrument (10X Genomics). Barcoding, cDNA amplification and library construction were performed using the Chromium Single Cell 3' reagent kits v.2. according to the manufacturer's instructions. Libraries were sequenced using the Novaseq 6000. Resulting fastqs were processed through CellRanger v2.2.0 using either the mm10 genome or an annotated *Acomys* genome to obtain a gene expression matrix. To help with downstream analyses *Acomys* genes were annotated with their mouse ortholog name if there was a one-to-one match, one gene from *Mus* matched one gene in *Acomys*.

Analyses were performed using Seurat v3^{155,156} and custom R scripts. Cells were filtered for cells containing over 500 genes, mitochondrial content below 20%, and hemoglobin below 2.5%. The mitochondrial and hemoglobin filtering was only performed in *Mus* samples because of missing annotation in *Acomys*. Only one read per cell was needed for a gene to be counted as expressed. The expression depth for each cell was normalized to 10000 genes per cell and log transformed¹⁵⁷. The data was scaled to reduce the effect of sequencing depth of each sample by regressing out this factor. When merging all data sets together, dimension reduction batch correction was performed by merging the data sets using 2000 anchor genes common to all data sets found within the FindIntegrationAnchors function and a final merging using the IntegrateData function. Dimension reduction was performed using the Python implementation of UMAP (Uniform Manifold Approximation and Projection) using genes common to both species. Cell clusters were determined by the Louvain algorithm. Marker genes were determined for each cluster using the Wilcoxon Rank Sum test within the FindAllMarkers function.

Putative signaling interactions between macrophages and fibroblasts, and macrophages and endothelial cells were examined. Potential receptor-ligand interactions were found by pairing a cell type expressing a ligand with a cell type expressing its receptor pair. A

receptor or ligand was considered expressed in a cell type if it had an average normalized expression of >0.2 . Receptor-ligand pairs were determined using the curated receptor-ligand database by the RIKEN FANTOM5 project¹⁵⁸. Receptor-ligand pairings for each cell type were visualized by a chord diagram using the R package *circlize*¹⁵⁹.

RNAscope—RNA *in situ* hybridization was performed using RNAscope® technology (Advanced cell diagnostics) (Wang et al., 2012). Ear pinna tissues fixed with 10% neutral buffered formalin for 16 hours at 4°C and washed with PBS and 70% ethanol. Tissue was embedded in paraffin and prepared in 5µm sections. A RNAscope Multiplex fluorescent Detection Reagent V2 kit (ACD, Cat. No. 323110) was used to hybridize the species-specific probes for *Cd14*, *Csf1r*, *Ltf*, *Vegfc* and *Pdgfra* on the section as per the manufacturer's protocol (Supplementary Table 16). Briefly, deparaffinized tissue sections were pretreated sequentially with hydrogen peroxide (10 min, at RT), RNAscope 1X Target retrieval buffer (15 min, at 100°C) and protease (30 min at 40°C). Pre-activated probes (10 min at 40°C) were hybridized to the samples at 40°C for 2 hours. The hybridized probes were amplified with Amp reagents. The probe signal was developed by sequential incubation with probe-specific HRP followed by TSA Plus fluorescent dyes (Akoya Biosciences) and HRP blocker. Nuclei were counter stained with Hoechst 33342. The sections were mounted with ProLong™ Gold antifade reagent (Invitrogen). Slides were imaged by epifluorescence microscope (Olympus IX83) at 60X magnification. RNA dots were counted per cell and plotted using JMP software.

QUANTIFICATION AND STATISTICAL ANALYSIS.

Statistical analysis for qPCR gene expression and Quansys protein expression were carried out using IBM SPSS Statistics 26 and GraphPad Prism9. Student t-test was run to analyze differences for baseline fibroblast gene expression *in vitro*. ANOVA was run to analyze differences in fibroblast gene expression before and after treatment with Macrophage conditioned media. Type III test for fixed effects was run for macrophage gene expression data before and after stimulation. A two-way ANOVA was run to analyze Quansys protein expression data, RNAscope quantification, and *Acomys* ear closure measurements which were followed by Post Hoc analysis using Tukey HSD or Student t-test. ANOVA was run to analyze specific protein effect on fibroblasts *in vitro*. Two-tailed Student's t-test run for all cell count data. Exact tests for each experiment are noted in the text or in Supplemental tables. Graphs were compiled in Graphpad Prism 9, excel or JMP Pro16. For RNAseq, *p*-values for gene expression fold changes were calculated using Cuffdiff.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Diversity and Inclusion Statement.

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in their field of research or within their geographical location. One or more of the authors of this paper self-identifies as a gender minority in their field of research. One or more of the authors of this paper received support from a program designed to increase minority representation in their field of research.

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Highlights

- *Acomys* macrophages exhibit a reduced inflammatory phenotype
- Secretory factors from *Acomys* macrophages induce regenerative fibroblast behavior
- Macrophage-secreted VEGFC facilitates complex tissue regeneration in *Acomys*

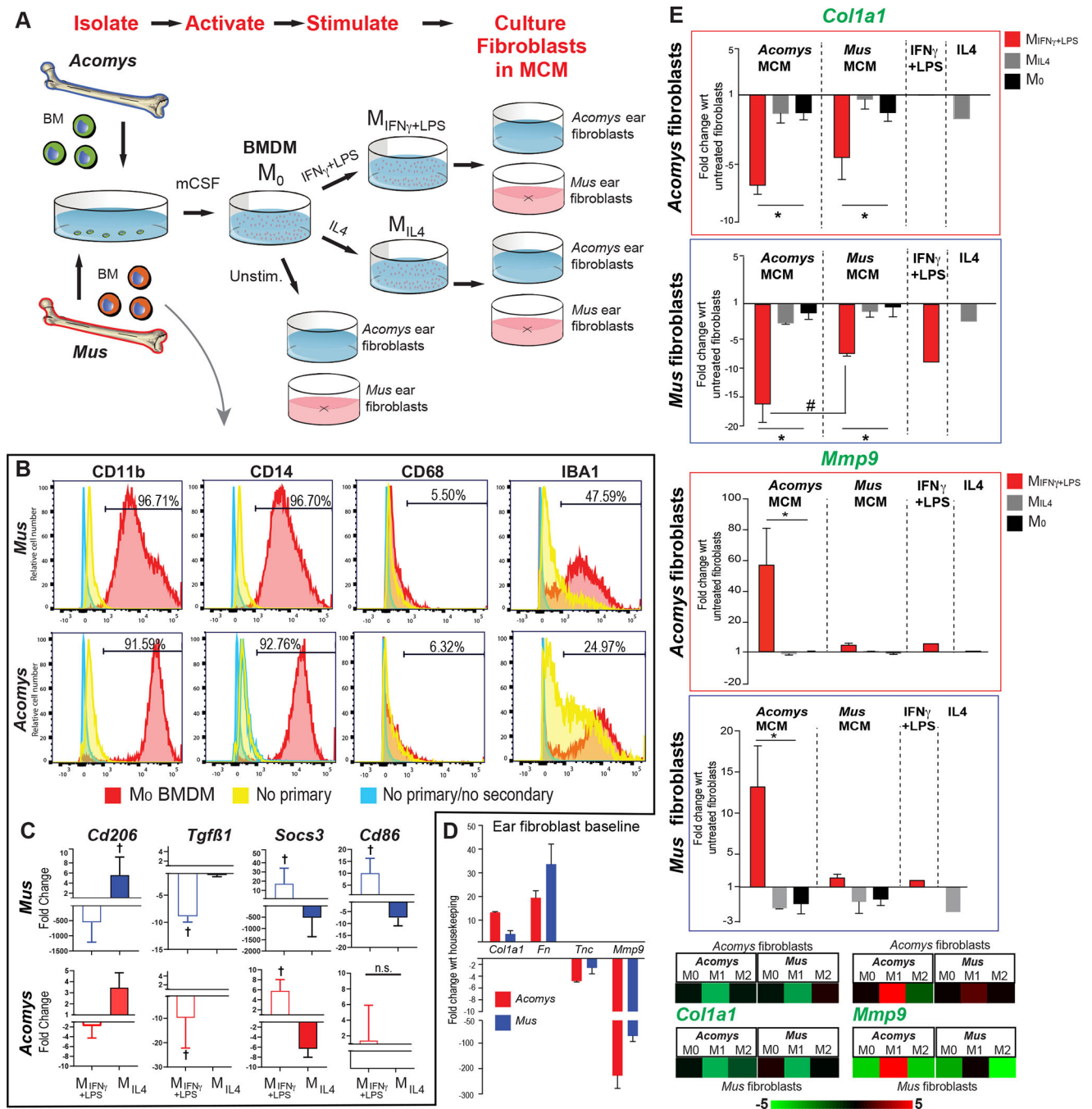


Figure 1. Conditioned media from *Acomys* bone marrow derived macrophages (BMDM) stimulated with IFN γ +LPS reduces collagen production and increases *Mmp9* expression in fibroblasts from both species.

A) Schematic of experimental design. Monocytes were isolated from *Acomys* or *Mus* bone marrow, activated with mCSF, then classically (IFN γ +LPS) or alternatively activated (IL4). These macrophages were cultured for 48hrs after which the media (macrophage conditioned media - MCM) was removed and used to stimulate *Mus* or *Acomys* primary ear fibroblasts. B) Flow analysis of BMDM using CD11b, CD14, CD68 and IBA1 after activation with mCSF and prior to stimulation with IFN γ +LPS or IL4. C) qPCR analysis for *Socs3*, *Cd86*,

Cd206 and *Tgf β 1* after stimulation with IFN γ +LPS ($M_{\text{IFN}\gamma+\text{LPS}}$) or IL4 (M_{IL4}). Gene expression presented as fold change over unstimulated cells (M_0). $\dagger p < 0.05$ for $M_{\text{IFN}\gamma+\text{LPS}}$ compared to M_{IL4} stimulation. D) qPCR analysis of *Acomys* or *Mus* ear fibroblasts before stimulation: *Collagen 1a1* (*Col1a1*), *Fibronectin* (*Fn*), *Tenascin C* (*TenC*), and matrix remodeling enzyme *Matrix Metalloprotease 9* (*Mmp9*). Fold change reported as target gene expression compared to housekeeping gene expression (*B2m* and *Tbp*). Mean $\pm$ SEM, $n=3$ (C-D). E) qPCR for *Col1a1* or *Mmp9* expression by *Mus* or *Acomys* ear fibroblasts after exposure to MCMs. Fold change reported as expression compared to untreated fibroblasts. Colored bars = gene expression in fibroblasts exposed to MCM from $M_{\text{IFN}\gamma+\text{LPS}}$ (red), M_{IL4} (grey) or M_0 (black) macrophages. $*p < 0.05$ for ANOVA comparing treatment within species. $\#p < 0.05$ for Tukey's multiple comparison test comparing responses to *Acomys* $M_{\text{IFN}\gamma+\text{LPS}}$ versus *Mus* $M_{\text{IFN}\gamma+\text{LPS}}$ (Table S3–4) Mean $\pm$ SEM, $n=3$. Bottom panel, heat map visualization of *Col1a1* and *Mmp9* expression.

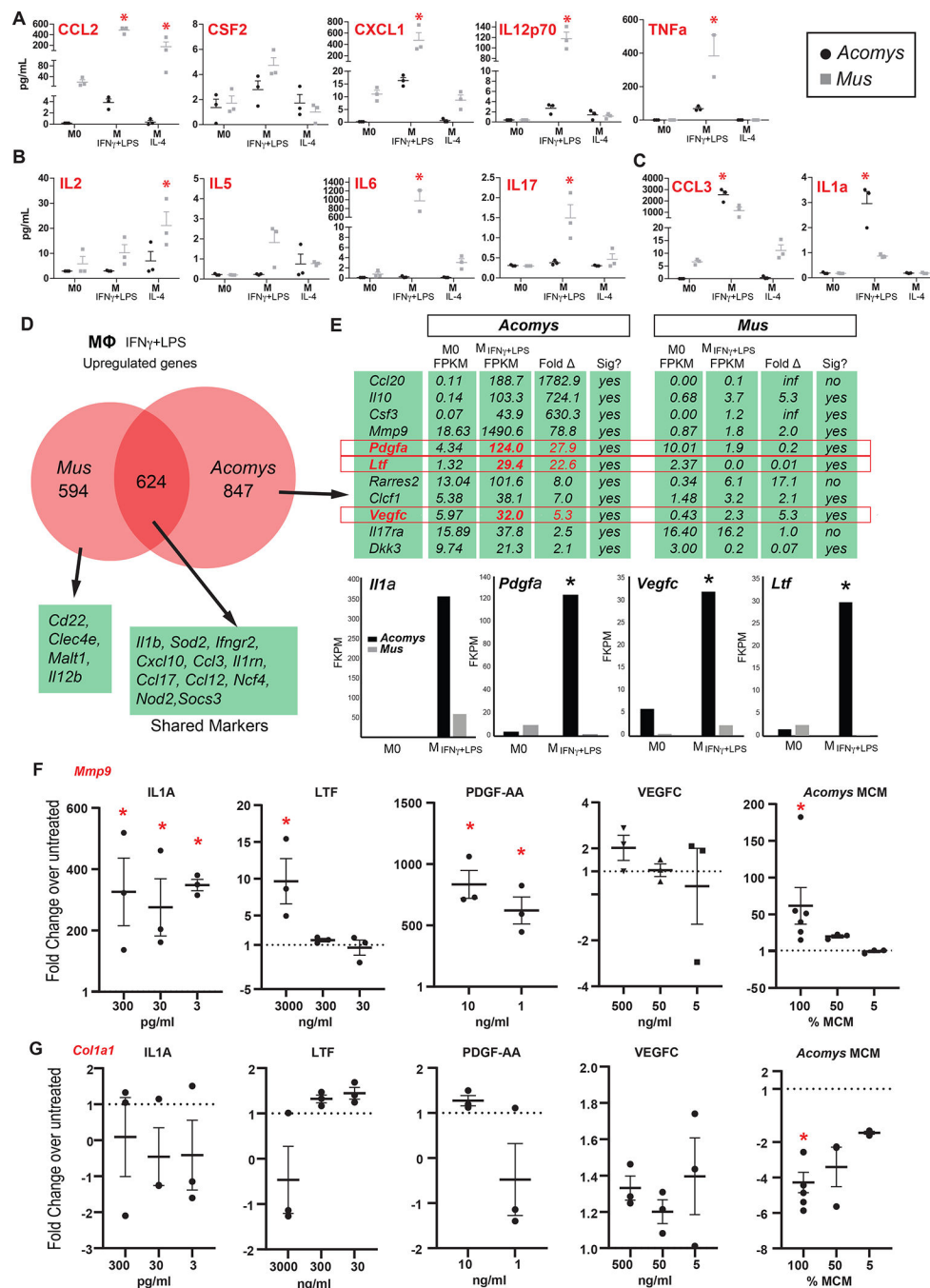


Figure 2. *Acomys* MIFN γ +LPS macrophages adopt a muted inflammatory profile compared to *Mus* macrophages and specifically express *Ltf*, *Vegfc* and *Pdgfa*.

A-C) Cytokine analysis of conditioned media (MCM) from *Acomys* and *Mus* bone marrow derived macrophages (M0) and from IFN γ +LPS and IL4 stimulation profiles. * $p < 0.05$ concentration (pg/ml) increase compared to M0. D) Venn diagram of upregulated genes in MIFN γ +LPS macrophages compared to M0 after RNA-seq analysis. Shared and unique macrophage markers depicted in boxes. E) Top 11 differentially expressed genes encoding secreted proteins that were (i) not expressed in M0 from either species, (ii) were significantly upregulated in *Acomys* MIFN γ +LPS macrophages and (iii) were not

expressed in *Mus*_{IFN γ +LPS} macrophages. FKPM = Fragments Per Kilobase of transcript per Million mapped reads. Increase in *Acomys* M_{IFN γ +LPS} macrophages compared to M₀ expression **p*<0.05. F-G) qPCR analysis for *Mmp9* (F) or *Col1a1* (G) expression by *Mus* ear fibroblasts stimulated with exogenous IL1A, LTF, PDGF-AA, VEGFC or *Acomys* macrophage conditioned media (MCM). Fold change represented as expression increase compared to unstimulated *Mus* ear fibroblasts. Mean $\pm$ SEM. **p*<0.05 compared to unstimulated fibroblasts.

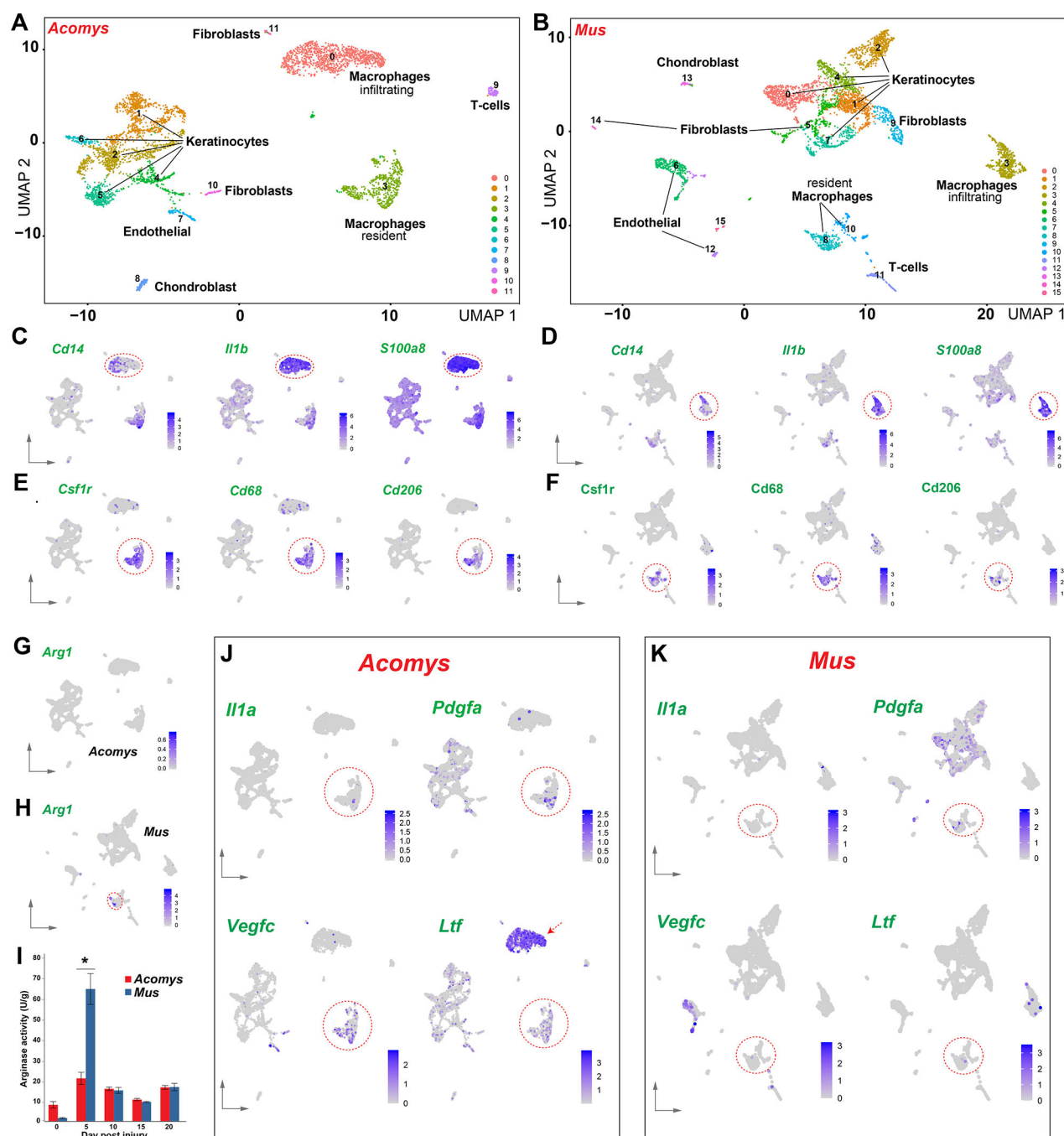


Figure 3. Single cell RNA-seq analysis of tissue undergoing regeneration or fibrotic repair five days post injury.

A-B) UMAP analysis of recovered cell types from healing ear pinna tissue at D5 (post injury) in *Acomys* (A) and *Mus* (B). C-F) Feature plots for genes enriched (purple) in *Acomys* (C) and *Mus* (D) infiltrating monocyte/macrophage (*Cd14*, *Il1b*, *S100a8*) genes and in *Acomys* (E) and *Mus* (F) tissue resident macrophage genes (*Csf1r*, *Cd68*, *Cd206*). G-H) Feature plot showing no *Arginase 1* (*Arg1*) expression in *Acomys* cells (G) and *Arg1* expression in *Mus* resident macrophages (H). I) Arginase 1 activity assay on tissue isolated at D0, 5, 10, 15, and 20 post injury from *Mus* (blue bars) or *Acomys* (red bars)

ear pinna. Two-way ANOVA, Sidak's multiple comparison test, $*p < 0.05$ increase in activity compared to other species at same time point. Mean $\pm$ SEM. J-K) *Il1a*, *Pdgfa*, *Vegfc*, and *Ltf* expression in *Acomys* (J) and *Mus* (J). Dotted red circles highlight infiltrating (C-D) or tissue resident (E-F) macrophage populations.

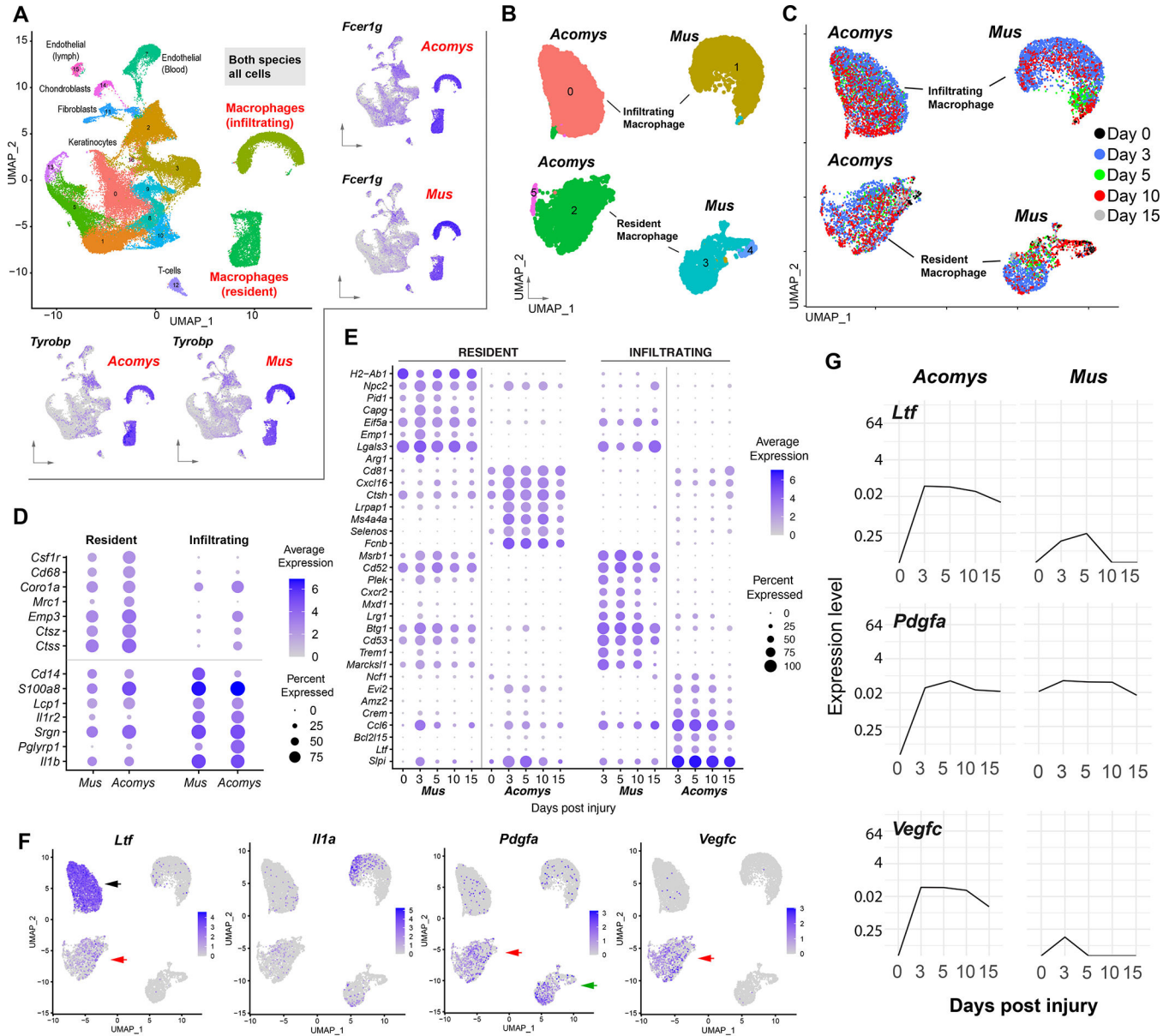


Figure 4. Resident and infiltrating macrophages maintain distinct identities throughout regeneration and fibrotic repair.

A) scRNA-seq analysis and UMAP projection of all cells from *Acomys* and *Mus* collected from tissue at D0, 3, 5, 10, and 15. Cells were batch-corrected to account for species-differences and segregated based on cell type. Feature plot analysis for *Tyrobp* and *Fcrl1g* which mark all macrophages in both species. B) UMAP for all macrophages at all timepoints separated by species and type. C) Day post injury mapped onto (B) with timepoints represented in legend. D) Top differentially expressed genes between infiltrating and resident macrophage clusters independent of species. E) Top differentially expressed genes separated by species and macrophage type (infiltrating vs. resident) across time. Point size represents percent cells expressing a specific gene. Color intensity represents average expression level for those cells expressing the gene of interest (D-E). F) Feature plots for *Ltf*, *Il1a*, *Pdgfa*, and *Vegfc*. G) Expression levels of *Ltf*, *Pdgfa*, and *Vegfc* over time.

Ltf, *Pdgfa*, *Il1a*, and *Vegfc* identified from our *in vitro* experiment (Fig. 3). Red (*Acomys*) and green (*Mus*) arrows highlighting resident macrophage population for all time points. Black arrow highlighting *Acomys* infiltrating population for all time points. G) Line graph of gene expression over time after injury for *Ltf*, *Pdgfa*, and *Vegfc* showing expression change over time.

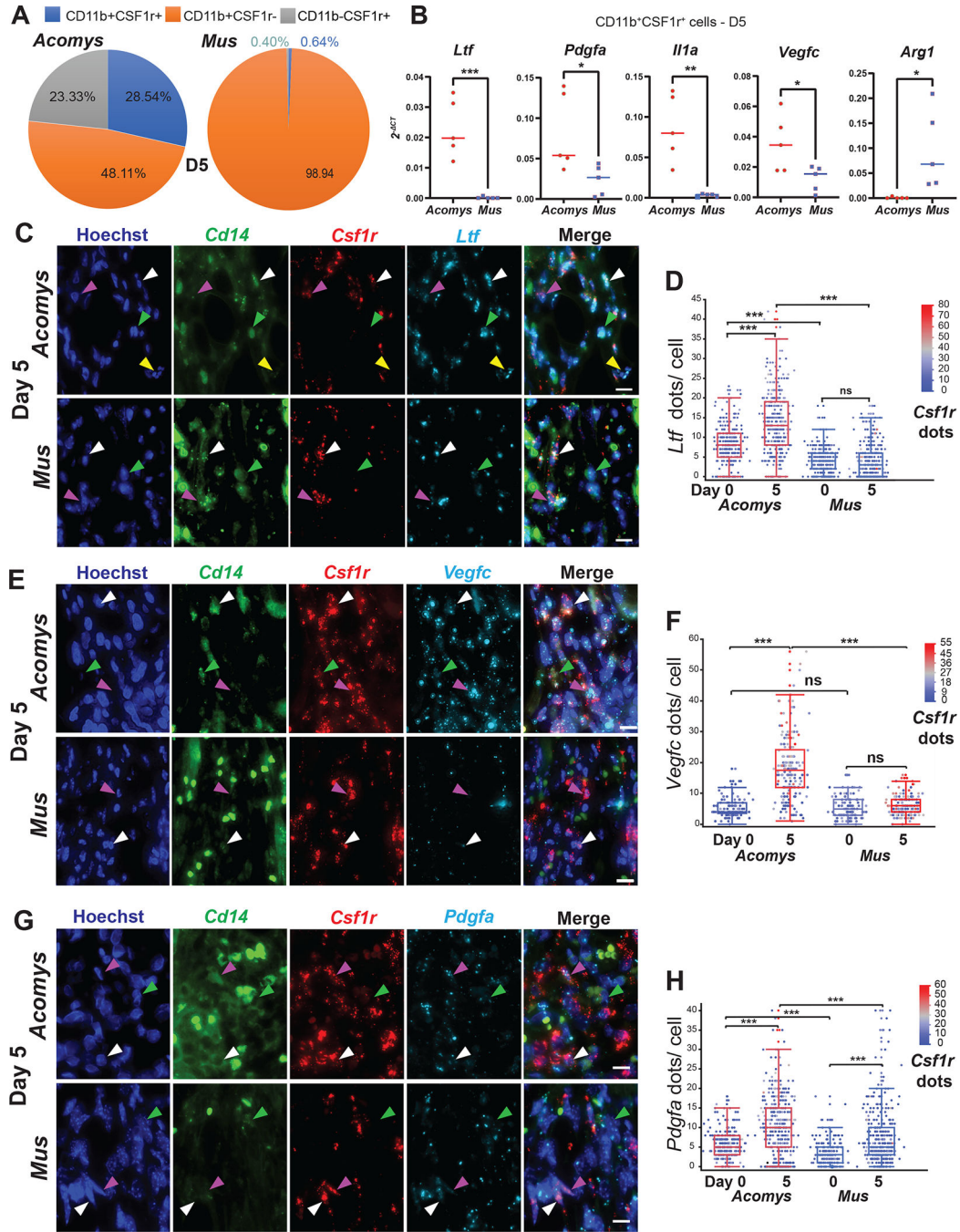


Figure 5. *Acomys* resident macrophages express high levels of *Vegfc* and *Ltf* transcripts five days post injury.

A) FACS analysis of macrophage population from ear pinna tissue at D5 using CD11b and CSF1r. B) Expression of *Ltf*, *Pdgfa*, *Il1a*, *Vegfc*, and *Arg1* in CD11b⁺;CSF1r⁺ cells isolated by FACS in (A). Fold change ($2^{-\Delta C_t}$) calculated as expression compared to housekeeping genes (*Tbp*, *B2m*). Two-tailed t-test * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. C-H) RNAscope analysis for *Cd14* and *Csf1r*, and either *Ltf*, *Vegfc* or *Pdgfa* expression at D5 post injury in *Mus* and *Acomys* ear pinna (C,E,G). Green arrow = *Cd14*⁺; *Csf1r*⁻; (*GOI*)⁺ cells where *GOI* = gene of interest. White arrow = *Cd14*⁺; *Csf1r*⁺; (*GOI*)⁺ cells. Purple arrow = *Csf1r*⁺; *Cd14*⁻; (*GOI*)⁺ cells.

(*GOL⁺*) cells. Green arrow = *Csf1r^{-/-};Cd14⁺*;*(GOL⁺)* cells. Yellow arrow = neutrophil. D) Quantification (fluorescent dots/cell) indicating number of transcripts per cell (D,F,H). Blue-red intensity legend represents number of *Csf1r gene* transcripts within the same cell. Scale bar = 10µm. Representative images from n=3 (D,F,H). Two-Way ANOVA and Post Hoc analysis by Tukey HSD. Mean ± SEM. **p*<0.05, ** *p*<0.01, ****p*<0.001, ns = not significantly different.

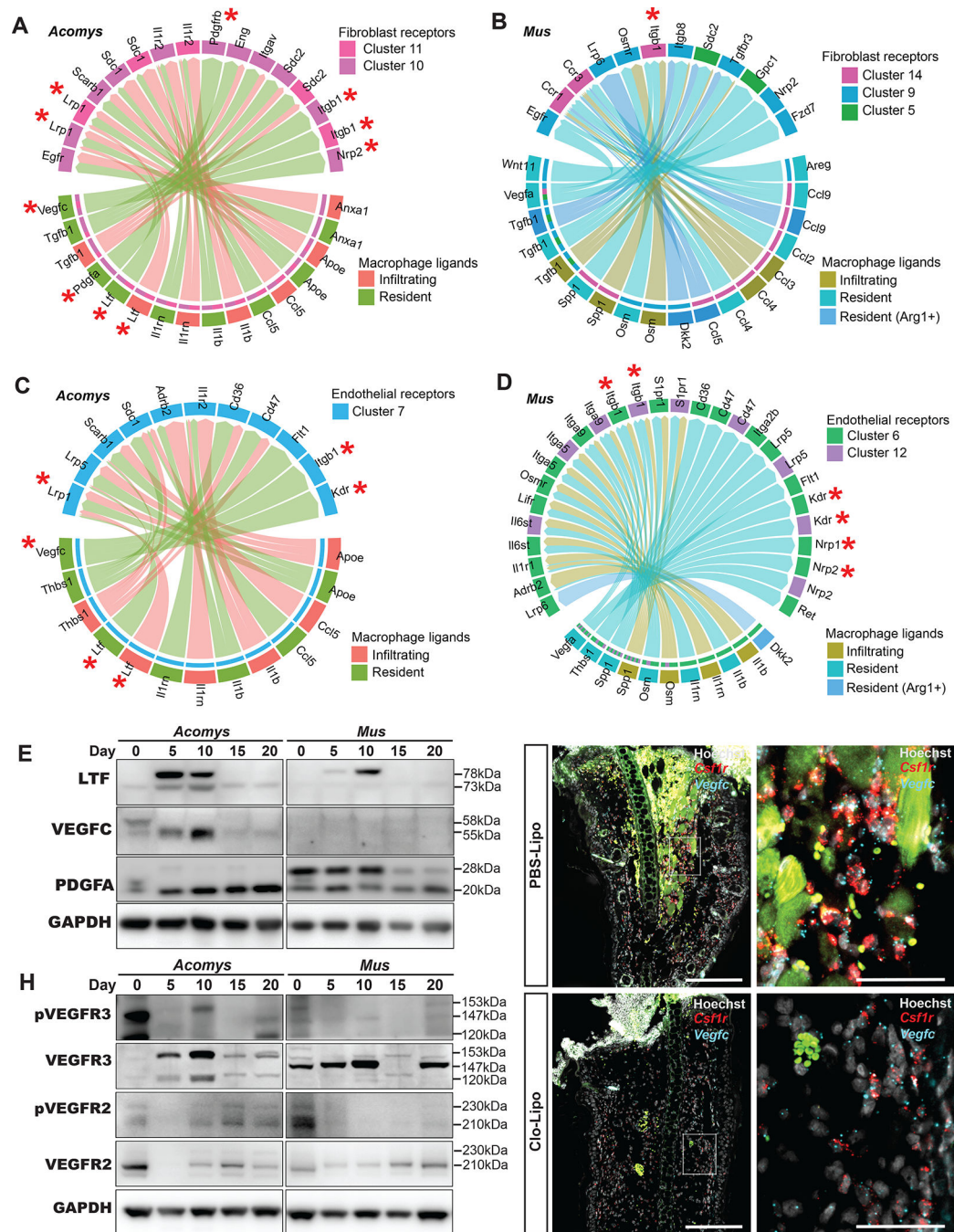


Figure 6. *Acomys* fibroblast and endothelial cells express receptors for *Vegfc* and *Ltf* ligands. A-D) Receptor-ligand analysis at D5 showing ligands expressed from infiltrating and resident macrophages from each species with cognate receptor expression for fibroblasts (A-B) and endothelial cells (C-D). Red asterisks mark *Vegfc-Itgb1* (*Acomys*), *Vegfc-Nrp2* (*Acomys*), *Ltf-Lrp1* (*Acomys*), *Pdgfa-Pdgfrb* (*Acomys*), and *Vegfa-Kdr* (*Mus*), *Vegfa-Nrp1* (*Mus*), *Vegfa-Nrp2* (*Mus*) and *Vegfa-Itgb1* (*Mus*). E) Western blot analysis of LTF, VEGFC and PDGFA proteins from ear pinna tissue isolated at D0, 5, 10, 15, and 20 post injury. Representative blots for n=3 animals. F-G) RNAscope analysis of *Csf1r* and *Vegfc*

expression at D5 post injury in PBS-Liposome (PBS-Lipo) and Clodronate Liposome (Clo-Lipo) injected *Acomys* ear pinna (see Methods). In response to Clo-Lipo, *Csf1r*⁺ and *Vegfc*⁺ cells were strongly reduced (G-G') compared to control ears (F-F'). Scale bar = 200µm (F, G), 50µm (F,G'). Representative images from n=3. H) Western blot analysis from ear pinna tissue isolated at D0, 5, 10, 15, and 20 post injury from *Mus* and *Acomys* for VEGFR2 and 3 and their phosphorylated forms at tyrosine 1175 and 1230 respectively. Representative blots for n=3 animals. GAPDH = loading control (E,H).

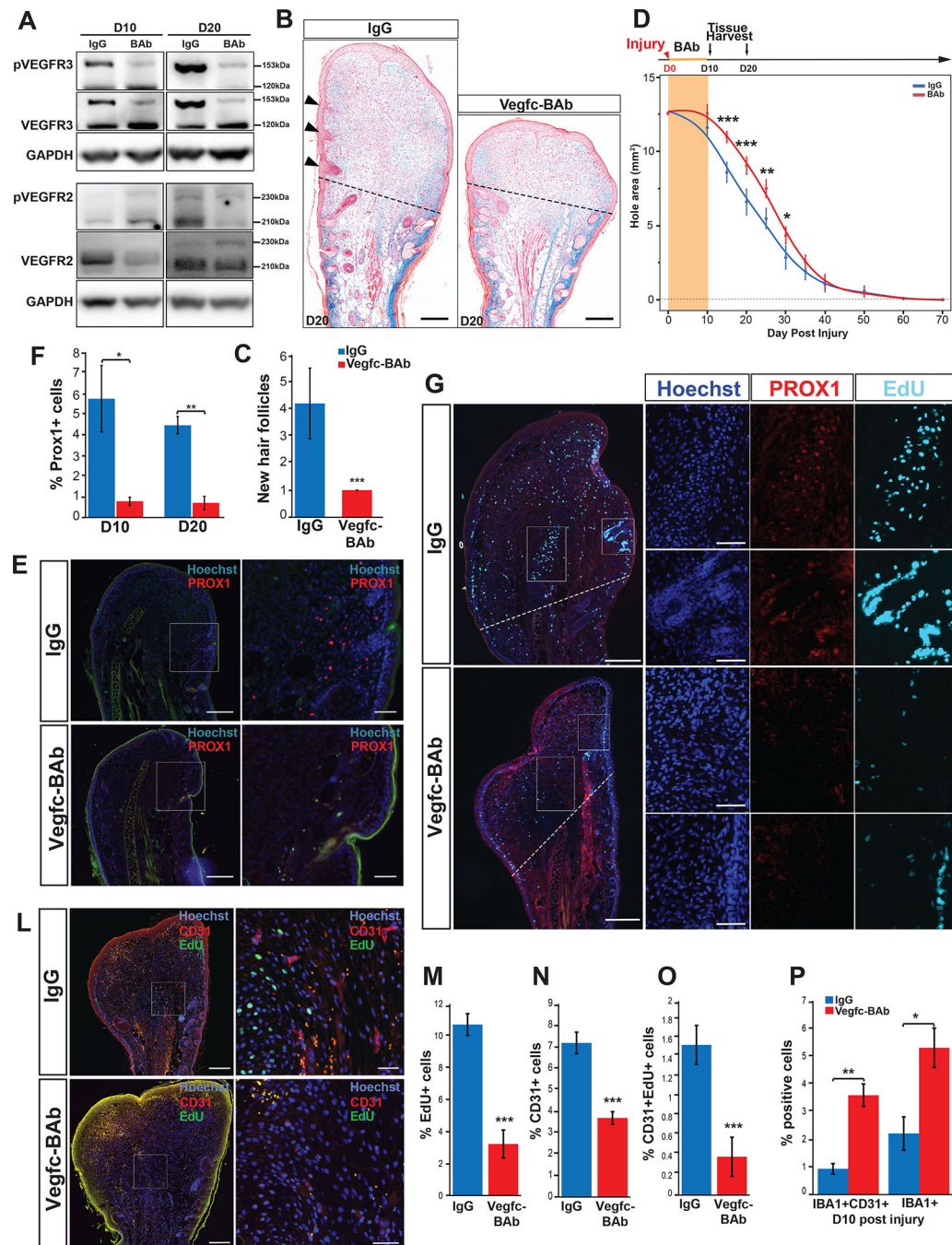


Figure 7. Inhibiting VEGFC-receptor interaction negatively impacts ear pinna regeneration. VEGFC blocking antibody (Vegfc-BAb) or Rabbit IgG (IgG) were injected subcutaneously daily into the base of the *Acomys* ear pinna for 10 days after injury. **A**) Western blot analysis of pVEGFR3, total VEGFR3, pVEGFR2, and total VEGFR2 at D10 and D20. GAPDH for loading control. Western analysis of pVEGFR3 and pVEGFR2 in Vegfc-BAb ears at D10 and D20 compared to control. **B**) Masson's trichrome stained sections of IgG and Vegfc-BAb at D20 with no newly forming hair follicles (black arrows). Injury site represented by black dotted line. **C**) Significantly fewer hair follicles in Vegfc-BAb ears

(n=4/group). D) Vegfc-Bab ears delayed new tissue formation, but ultimately open holes closed by D60 (n=5/group). Two-Way ANOVA and Post Hoc analysis by Student's t-test pairwise comparison. E) Immunofluorescence for PROX1 at D10. Magnified regions in E to the right panel show significantly fewer PROX1⁺ cells in treatment ears. F) PROX1⁺ cells (from E) significantly declined in Vegfc-BAB treated ears. G-K) Immunofluorescence for PROX1 and EdU at D20 in IgG or Vegfc-BAB treated ears. Boxed regions are magnified from treatment and controls and shown in separate channels for IgG (H-I) and Vegfc-Bab (J-K) (n=5/group). L) Immunostaining for CD31 and EdU at D20 in Vegfc-Bab and control ears. Boxed regions highlight decline in CD31⁺ cells. M-O) Quantification of percent EdU⁺ (M), CD31⁺ (N) and EdU⁺;CD31⁺ (O) cells normalized to total Hoechst positive cells at D20 (n=5/group). P) IBA1⁺ macrophages with and without CD31 in IgG or Vegfc-BAB treated at D10 (n=4/group). Scale bars = 200µm (C,E,G,L) and 50µm (E,H-K,L insets). Mean ± SEM. Two tailed Student's t-test in C, F, M-P. **p*<0.05, ** *p*<0.01, ****p*<0.001.

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rat anti-mouse F4/80	BD Biosciences	Clone T45–2342
Goat anti-mouse IBA1	Novus Biologicals	Cat#NB100–1028
PE-Conjugated CD11b	BD Pharmingen	Cat#557397
APC-conjugated CD115	BioLegend	Cat#347306
Rabbit anti CD14	Proteintech	Cat#17000–1-AP
Mouse anti CD68	Abcam	Cat#ab955
Rabbit anti CD11b	Novus Biologicals	Cat#NB110–89474
Rabbit anti Vegfc	Novus Biologicals	Cat#NB110–61022
Rabbit anti Pdgfa	Novus Biologicals	Cat#NBP2–99113
Anti Phospho-VEGFR3	Biorbyt	Cat#orb1095455
Anti VEGFR3	Biorbyt	Cat#orb315557
Anti Phospho-VEGFR2	Cell Signaling Technology	Cat#2478
Anti VEGFR2	Cell Signaling Technology	Cat#2479
Anti GAPDH	Cell Signaling Technology	Cat#2118
Anti LTF	Invitrogen	Cat#PA5–95513
Anti α SMA	Novus Biologicals	Cat#NB300–978
Anti CD31	R&D systems	Cat#AF3628
Anti LYVE1	Invitrogen	Cat#13044382
Anti PROX1	AngioBio	Cat#11–002P
Rabbit VEGFC-C blocking antibody	ANGIO-PROTEOMIE	Cat#pV1006R-r
Rabbit IgG control	Vector Laboratories	Cat#I-1000
Mouse IgG control	Vector Laboratories	Cat#I-2000
Anti rabbit-HRP	Santa Cruz Biotechnology	Cat#sc-2030
Chemicals, Peptides, and Recombinant Proteins		
LPS from E.coli	Sigma	Cat#L3012
recombinant mouse IFN γ	Peptotech	Cat#315–05
recombinant mouse IL4	Peptotech	Cat#214–14
recombinant human IL1A	Peptotech	Cat#200–01A
recombinant mouse VEGFC	BioLegend	Cat#775104
recombinant human PDGF-AA	Peptotech	Cat#100–13A
PerfeCTA SYBR Green	QuantaBio	Cat#95054
Hypothermosol FRS	Sigma	Cat#H4416
Protease from Bacillus Licheniformis	Sigma	Cat#P5380
complete™, Mini, EDTA-free Protease Inhibitor Cocktail	MilliporeSigma	Cat#11836170001
Simple Stop™ 2 Phosphatase Inhibitor Cocktail	GOLDBIO	Cat#GB-451–1
EdU (5-ethynyl-2'-deoxyuridine)	Invitrogen	Cat#A10044

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical Commercial Assays		
RNAscope Multiplex Fluorescent Detection Reagent V2 kit	Advanced Cell Diagnostics	Cat#323110
Sandwich ELISA array	Quansys Biosciences	custom kit
Chromium Single Cell 3' reagent kits v.2	10x Genomics	Cat#PN-120237
SensiFAST cDNA synthesis kit	Thomas Scientific	Cat#C755H65
BCA assay kit	Pierce, Thomas Scientific	Cat#23225
Clarity TM ECL detection kit	BIO-RAD	Cat#170 –5060
Deposited Data		
<i>Mus</i> mm10 genome	http://hdownload.soe.ucsc.edu/downloads.html#mouse	NA
scRNA-seq data	N/A	GSE182141
Experimental Models: Cell Lines		
L929 cells	ATCC	CCL1
Experimental Models: Organisms/Strains		
<i>Mus musculus</i> : Swiss Webster	Envigo	Hsd:Nd4
<i>Acomys cahirinus</i>	Bred in house at University of Kentucky	N/A
Oligonucleotides		
RNAscope Mouse <i>Cd14</i>	Advanced Cell Diagnostics	Cat#451061
RNAscope Mouse <i>Csf1r</i>	Advanced Cell Diagnostics	Cat#428191-C2
RNAscope Mouse <i>Ltf</i>	Advanced Cell Diagnostics	Cat#823571-C3
RNAscope Mouse <i>Pdgfa</i>	Advanced Cell Diagnostics	Cat#411361-C3
RNAscope Mouse <i>Vegfc</i>	Advanced Cell Diagnostics	Cat#492701-C3
RNAscope Acomys <i>Cd14</i>	Advanced Cell Diagnostics	Cat#849831
RNAscope Acomys <i>Csf1r</i>	Advanced Cell Diagnostics	Cat#849851-C2
RNAscope Acomys <i>Ltf</i>	Advanced Cell Diagnostics	Cat#849801-C3
RNAscope Acomys <i>Pdgfa</i>	Advanced Cell Diagnostics	Cat#849761-C3
RNAscope Acomys <i>Vegfc</i>	Advanced Cell Diagnostics	Cat#849791-C3
Primers for quantitative PCR	see Table S16 for full sequences	N/A
Software and Algorithms		
Tophat2	http://ccb.jhu.edu/software/tophat/index.shtml	10.1186/gb-2013-14-4-r36
Cufflinks	http://cole-trapnell-lab.github.io/projects/cufflinks/	doi:10.1038/nbt.1621
Cell Ranger v2.2.0	https://support.10xgenomics.com/single-cell-gene-expression/software/release-notes/2-0	N/A
Seurat v3	https://github.com/satijalab/seurat	10.1016/j.cell.2019.05.031
Fiji	Schindelin et al. ¹⁴⁵	https://fiji.sc
FloJo software (version 10)	N.A.	https://www.flowjo.com/solutions/flowjo/downloads